

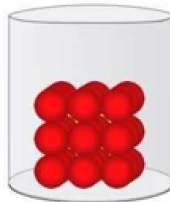
Solid State

States of Matter

Matter

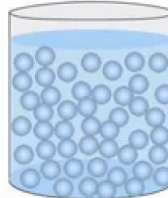
3 States

solid



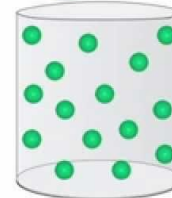
- rigid
- fixed shape
- fixed volume

liquid



- not rigid
- no fixed shape
- fixed volume

gas



- not rigid
- no fixed shape
- no fixed volume

- Liquids & Gases

Fluids

Ability to flow



- Solids

Rigids



- At given set of T & P , **Most stable state** of substance depends on -

1. Intermolecular forces



Tend to keep the molecules
(atoms and ions) closer

2. Thermal energy

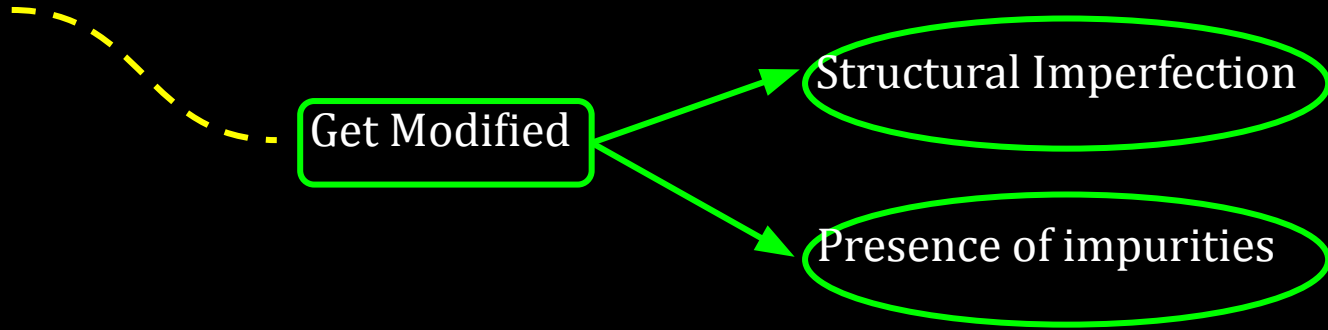


Tends to keep molecules
(atoms and ions) apart

- At low T** , thermal energy is low and intermolecular forces bring molecules(atoms and ions) very close. Hence, **substance exists in solid state**.

General Characteristics of Solids

- Structures of solids : Due to different arrangement of solid particles.
- Properties of solids : Due to different structures of solids

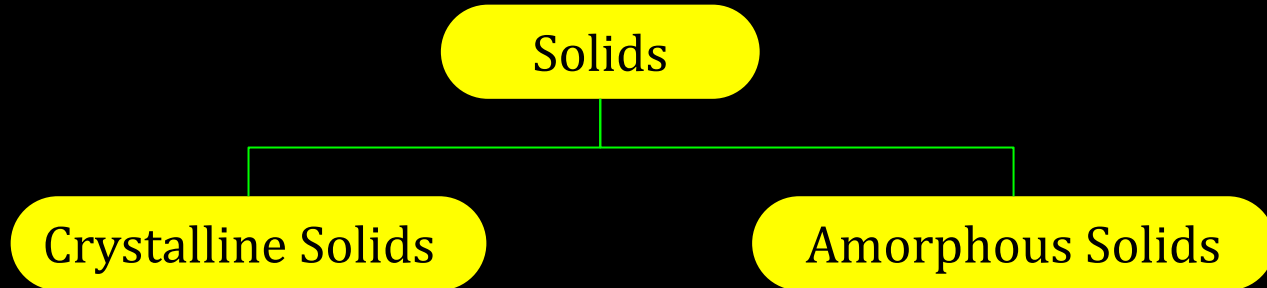


The following are the characteristic properties of the solid state:

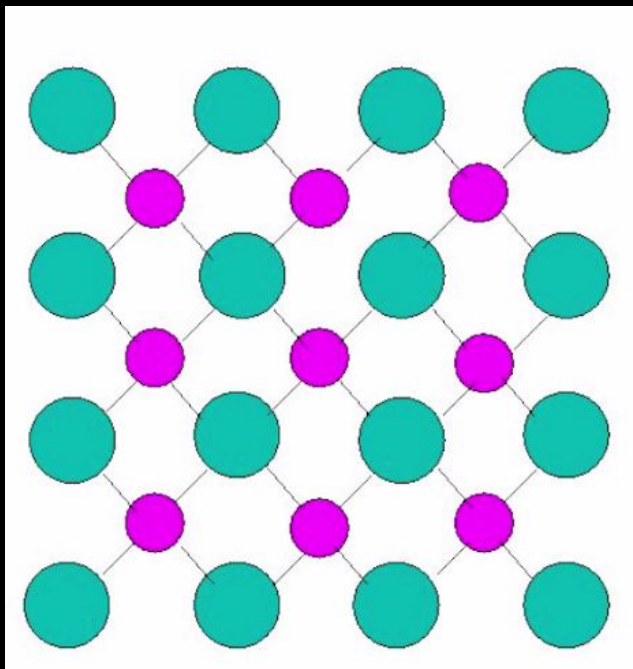
- They have definite mass, volume and shape.
- Intermolecular distances are short.
- Intermolecular forces are strong.
- They are incompressible and rigid.

Types of Solids :

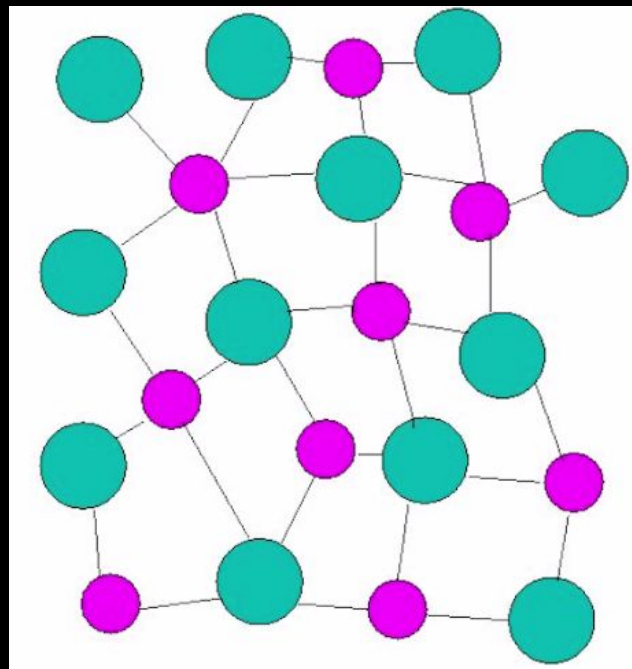
On the Basis of nature of order present in the arrangement of particles :



Crystalline Solids



Amorphous Solids



Crystalline Solids

Long range order

Definite Shape

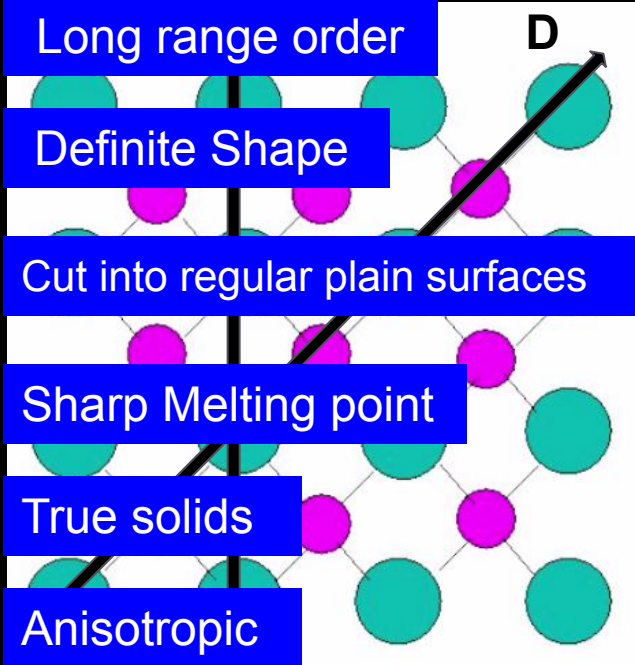
Cut into regular plain surfaces

Sharp Melting point

True solids

Anisotropic

D



Amorphous Solids

Short range order

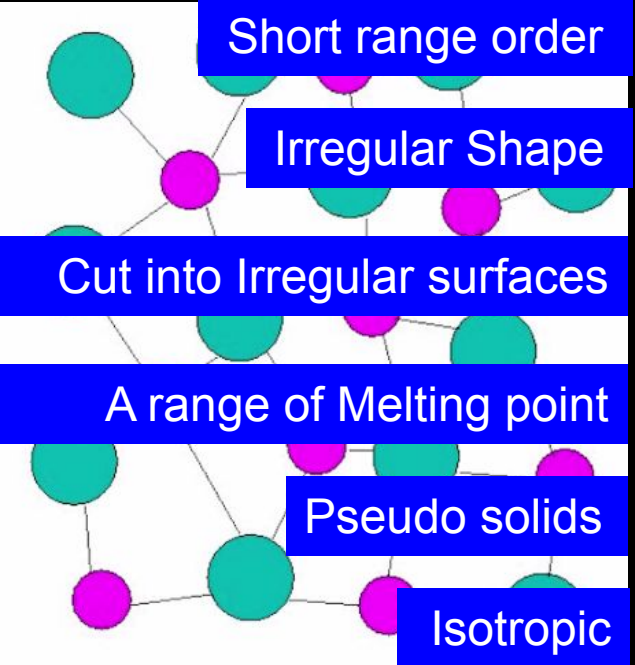
Irregular Shape

Cut into Irregular surfaces

A range of Melting point

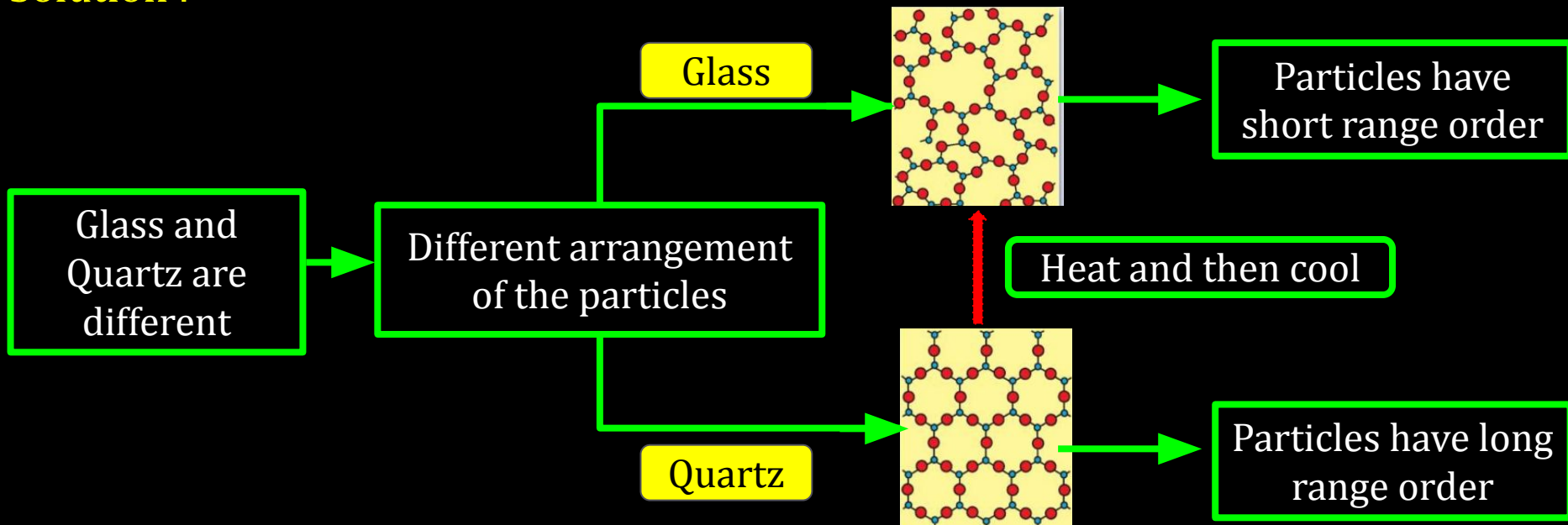
Pseudo solids

Isotropic



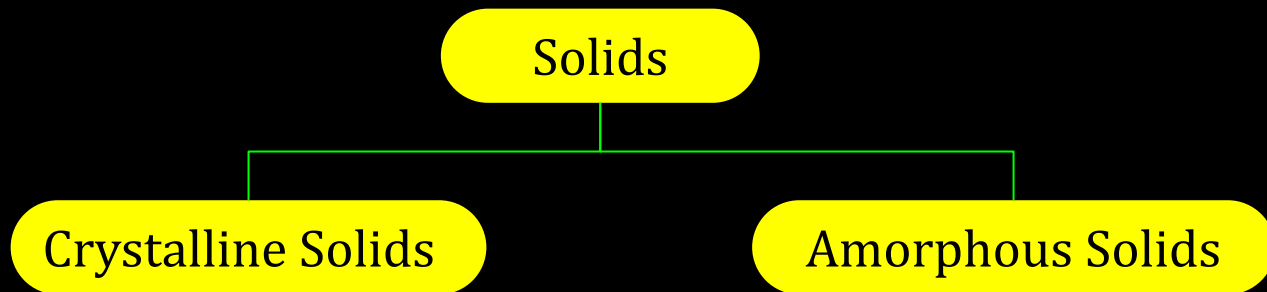
Question : What makes a glass different from a solid such as quartz? Under what conditions could quartz be converted into glass?

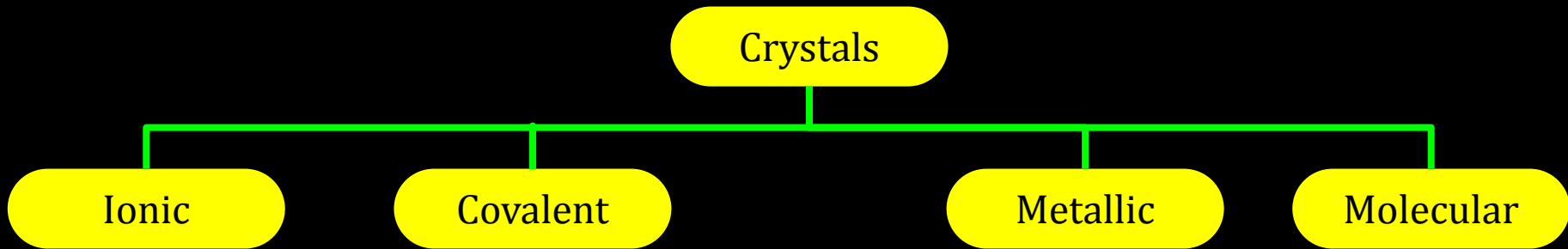
Solution :



Types of Solids :

On the Basis of nature of order present in the arrangement of particles:





Crystals

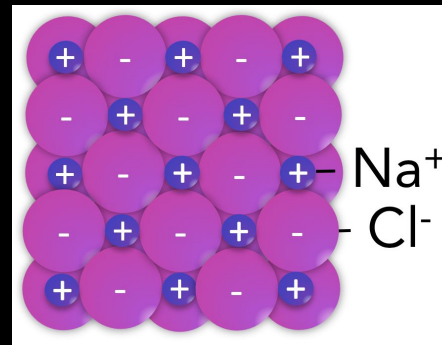
Ionic

Covalent

Metallic

Molecular

- Made of Ions
- Electrostatic forces
- Hard and Brittle
- High melting and boiling points
- Electrical insulators in solid state
- Electrical conductors in molten and aqueous state.
- Examples are NaCl, CaF_2 .



Non-polar

Polar

H - bonded

Crystals

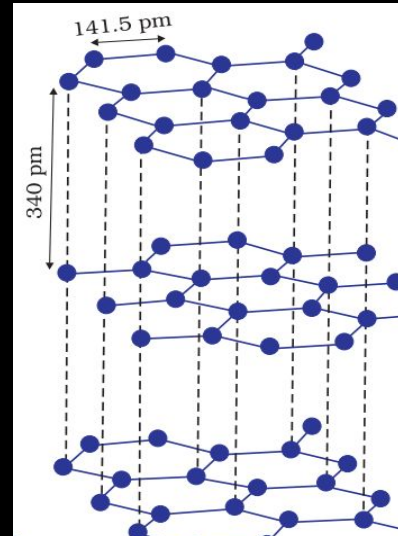
Ionic

Covalent

Metallic

Molecular

- Covalent bonds are present.
- Such solids are very hard and brittle.
- Extremely high melting points.
- They are insulators and do not conduct electricity.
- Examples are, diamond, silicon carbide and graphite.
- Exceptionally, Graphite is soft and is a conductor of electricity.



Non-polar

Polar

H - bonded

Crystals

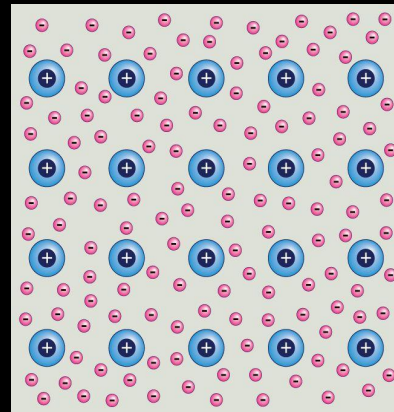
Ionic

Covalent

Metallic

Molecular

- Metallic bonds are present.
- Collection of positive ions surrounded by and held together by a sea of free electrons.
- High electrical and thermal conductivity due to free electrons.
- Lustrous, malleable and ductile in nature.
- Example are Fe, Cu, Ag etc.



Non-polar

Polar

H - bonded

Crystals

Ionic

Covalent

Metallic

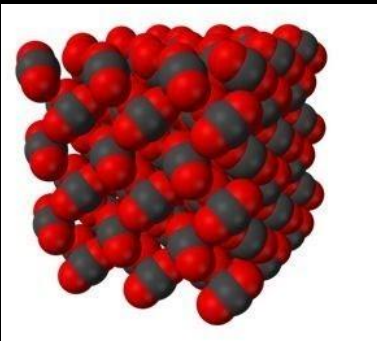
Molecular

- Molecules are the constituent particles of molecular solids.

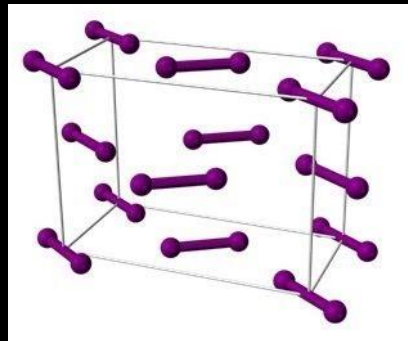
Non-polar

Polar

H - bonded



Carbon di oxide
(CO₂)



Iodine
(I₂)

12C01.1 Solid State & its Classification

Crystals

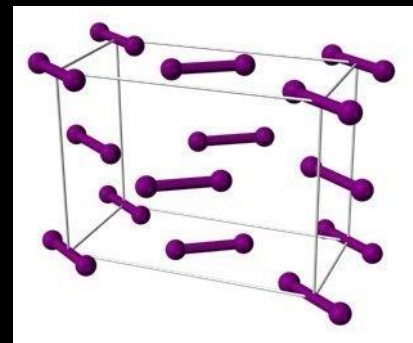
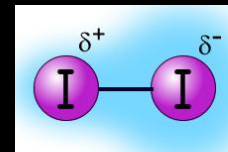
Ionic

Covalent

Metallic

Molecular

- Contain either atoms like He and Ar or non polar molecules like H_2 , Cl_2 and I_2 .
- Weak dispersion or London forces are present.
- Soft and non-conductors of electricity.
- Low melting points
- liquid or gaseous state at room temp .



Iodine (I_2)

Non-polar

Polar

H - bonded

12C01.1 Solid State & its Classification

Crystals

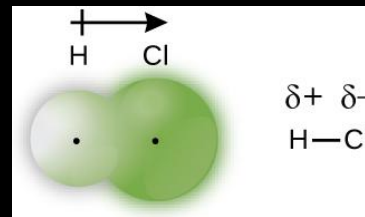
Ionic

Covalent

Metallic

Molecular

- Formed by molecules like HCl , SO_2 with Polar covalent bonds.
- Strong dipole-dipole interactions are present.
- Soft and non-conductors of electricity.
- M.P are higher than non polar molecular solids.
- Generally, gases or liquids at room temperature.



Non-polar

Polar

H - bonded

12C01.1 Solid State & its Classification

Crystals

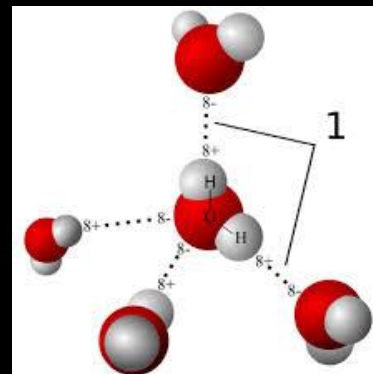
Ionic

Covalent

Metallic

Molecular

- Molecules of these solids contain polar covalent bonds between H and F, O or N atoms.
- Hydrogen bonds are present.
- Non-conductors of electricity.
- Volatile liquids or soft solids under room temp.
- Example is H_2O (ice).



Non-polar

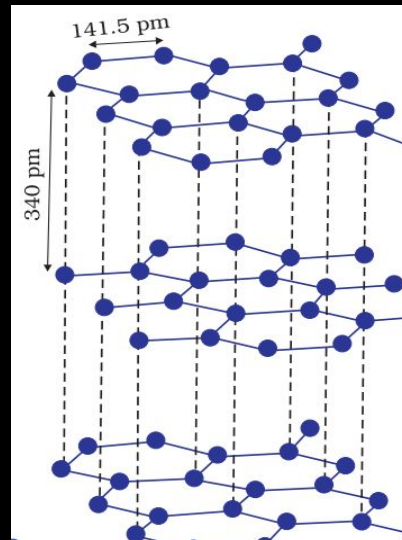
Polar

H - bonded

Graphite - Exception of Covalent Or Network Solid

Exceptional properties of graphite are due to its **typical structure**.

- C- atoms are arranged in **different layers**.
- Each atom is **covalently bonded** to three of its neighbouring atoms in same layer.
- **Fourth valence electron** of each atom is present between different layers and is **free to move about**.
- Free electrons make it a **good conductor**.



Question : Classify the following as ionic, metallic, molecular, covalent or amorphous.

- (i) Tetra phosphorus decoxide (P_4O_{10}) (ii) Ammonium phosphate $(NH_4)_3PO_4$
 (iii) SiC (iv) I_2 (v) P_4 (vi) Plastic (vii) Graphite (viii) Brass (ix) Rb (x) LiBr (xi) Si

Solution :

Ionic

(ii) Ammonium phosphate $(NH_4)_3PO_4$, (x) LiBr

Metallic

(viii) Brass, (ix) Rb

Molecular

(i) Tetra phosphorus decoxide (P_4O_{10}), (iv) I_2 , (v) P_4

Covalent

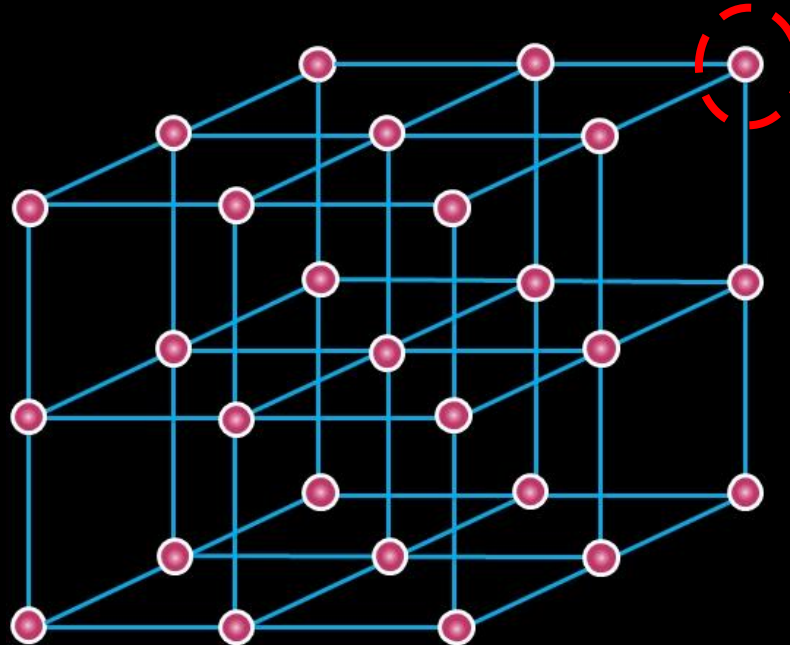
(iii) SiC, (vii) Graphite, (xi) Si

Amorphous

(vi) Plastic

- **Crystal Lattice, Unit Cell and its Parameters**
- **Types of Unit Cells**
- **Bravais Lattices**
- **Number of Atoms in a Cubic Unit Cell**

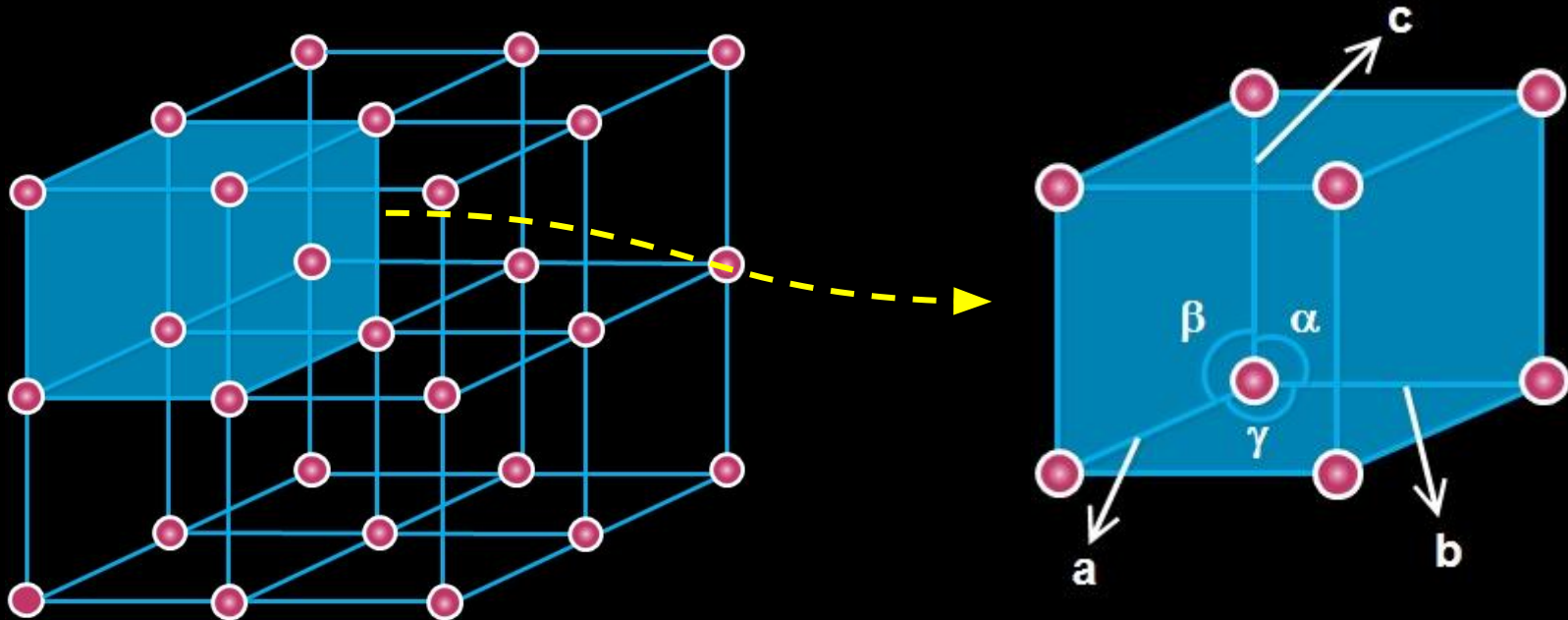
Crystal lattice or space lattice



Each point called
Lattice Point

Atom,
molecule or ion

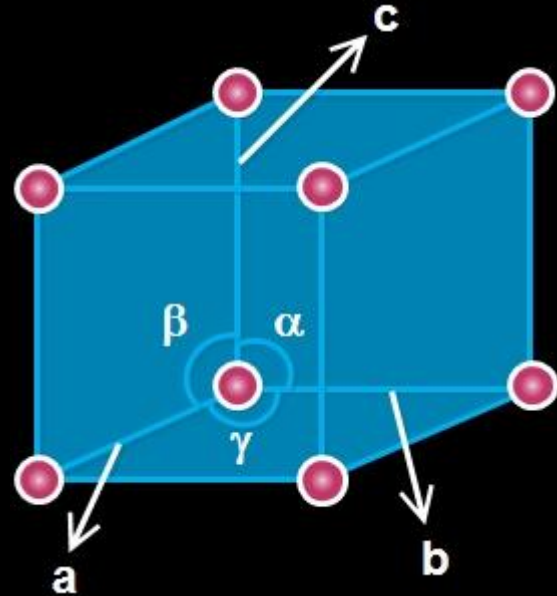
Unit cell



Parameters of unit cell
- 6 parameters

Edge lengths a , b and c .

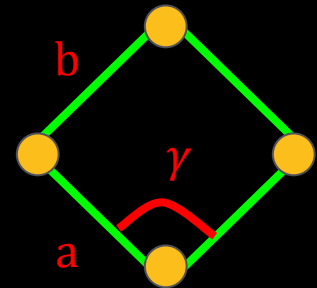
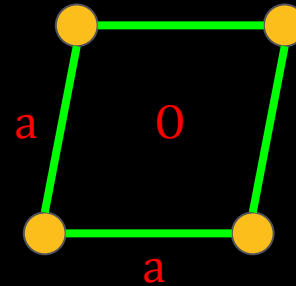
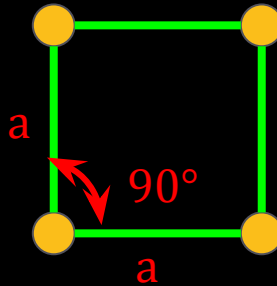
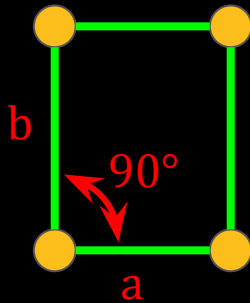
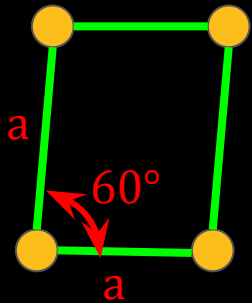
Angles between the edges :
 α , β , γ

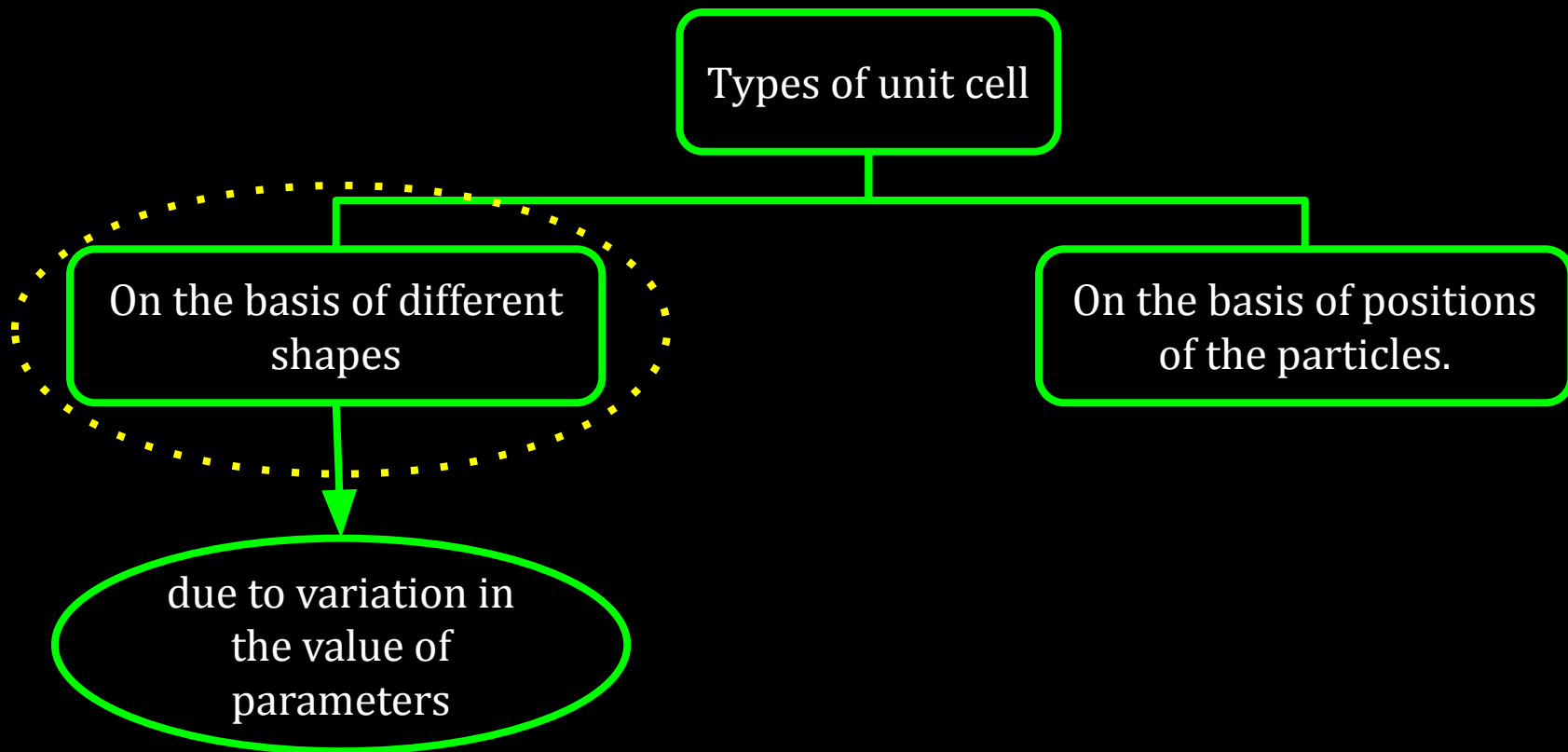


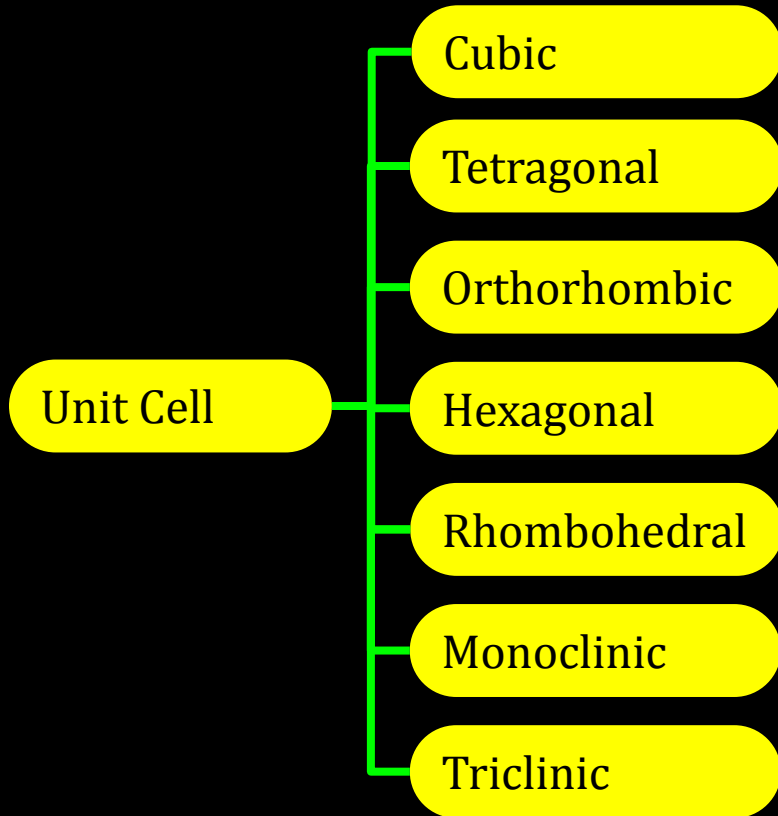
Unit Cells in 2-D :

In **two dimensions** a parallelogram with side of length '**a**' and '**b**' and an **angle γ** between these sides is chosen as unit cell.

Possible unit cells in two dimensions







Unit Cell

Cubic

Tetragonal

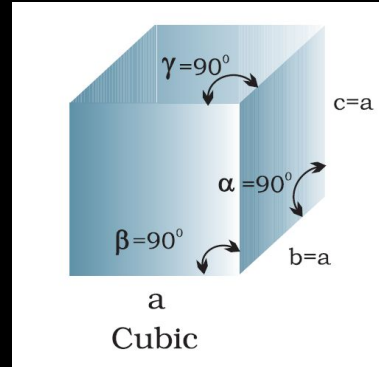
Orthorhombic

Hexagonal

Rhombohedral

Monoclinic

Triclinic



Edge lengths: $a = b = c$

Axial angles:
 $\alpha = \beta = \gamma = 90^\circ$

Examples: NaCl, Zinc
blend, Cu

Unit Cell

Cubic

Tetragonal

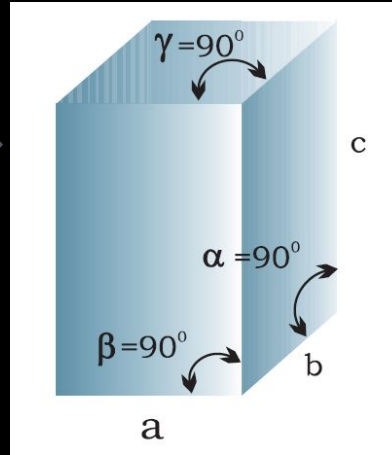
Orthorhombic

Hexagonal

Rhombohedral

Monoclinic

Triclinic



Edge lengths: $a = b \neq c$

Axial angles:
 $\alpha = \beta = \gamma = 90^\circ$

Examples: White tin,
 SnO_2 , TiO_2 , CaSO_4

Unit Cell

Cubic

Tetragonal

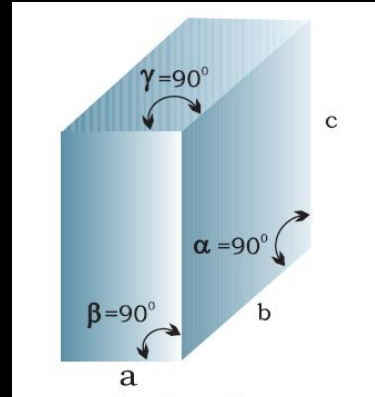
Orthorhombic

Hexagonal

Rhombohedral

Monoclinic

Triclinic



Edge lengths: $a \neq b \neq c$

Axial angles:
 $\alpha = \beta = \gamma = 90^\circ$

Examples: Rhombic sulphur, KNO_3 , BaSO_4

Unit Cell

Cubic

Tetragonal

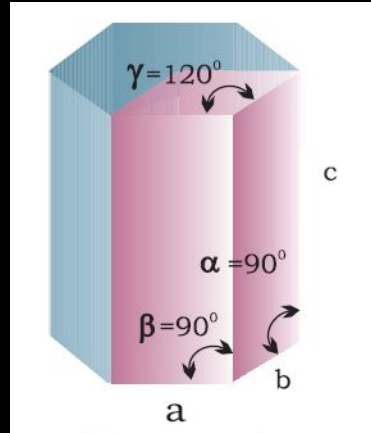
Orthorhombic

Hexagonal

Rhombohedral

Monoclinic

Triclinic



Edge lengths: $a = b \neq c$

Axial angles:

$$\alpha = \beta = 90^\circ$$

$$\gamma = 120^\circ$$

Examples: Graphite, ZnO, CdS

Unit Cell

Cubic

Tetragonal

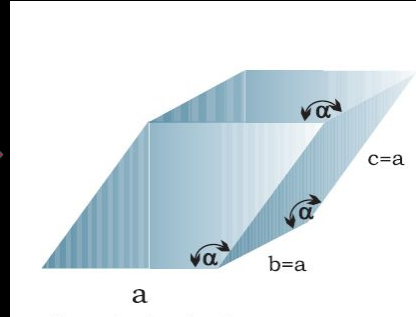
Orthorhombic

Hexagonal

Rhombohedral

Monoclinic

Triclinic



Edge lengths: $a = b = c$

Axial angles:
 $\alpha = \beta = \gamma \neq 90^\circ$

Examples: Calcite (CaCO_3), HgS (cinnabar)

Unit Cell

Cubic

Tetragonal

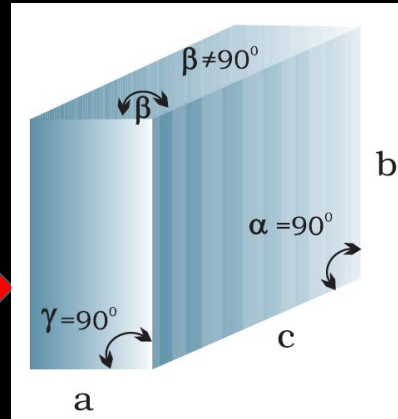
Orthorhombic

Hexagonal

Rhombohedral

Monoclinic

Triclinic



Edge lengths: $a \neq b \neq c$

Axial angles:

$$\alpha = \gamma = 90^\circ$$

$$\beta \neq 90^\circ$$

Examples: Monoclinic sulphur, $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$

Unit Cell

Cubic

Tetragonal

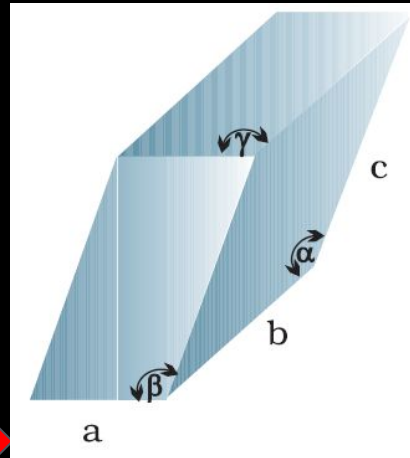
Orthorhombic

Hexagonal

Rhombohedral

Monoclinic

Triclinic

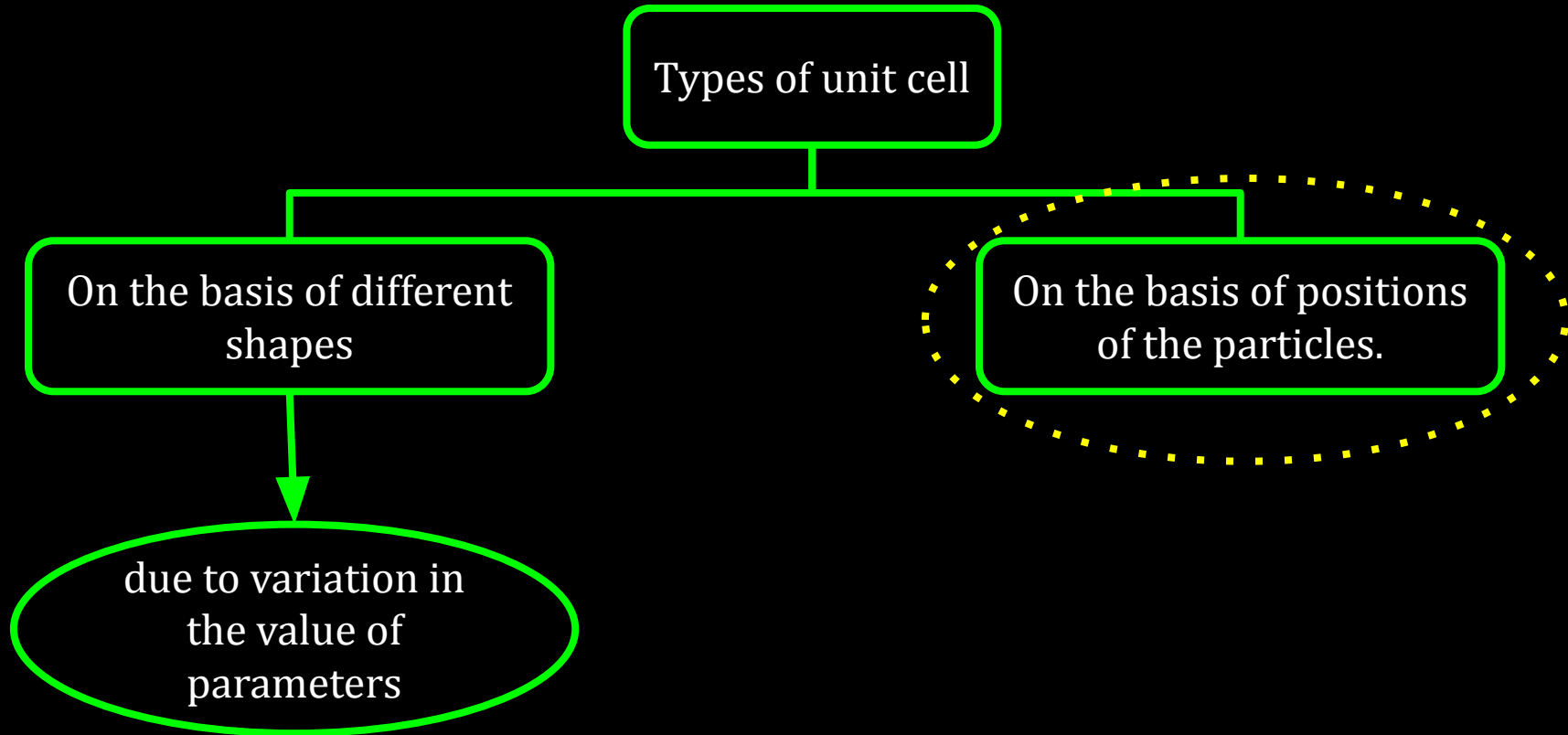


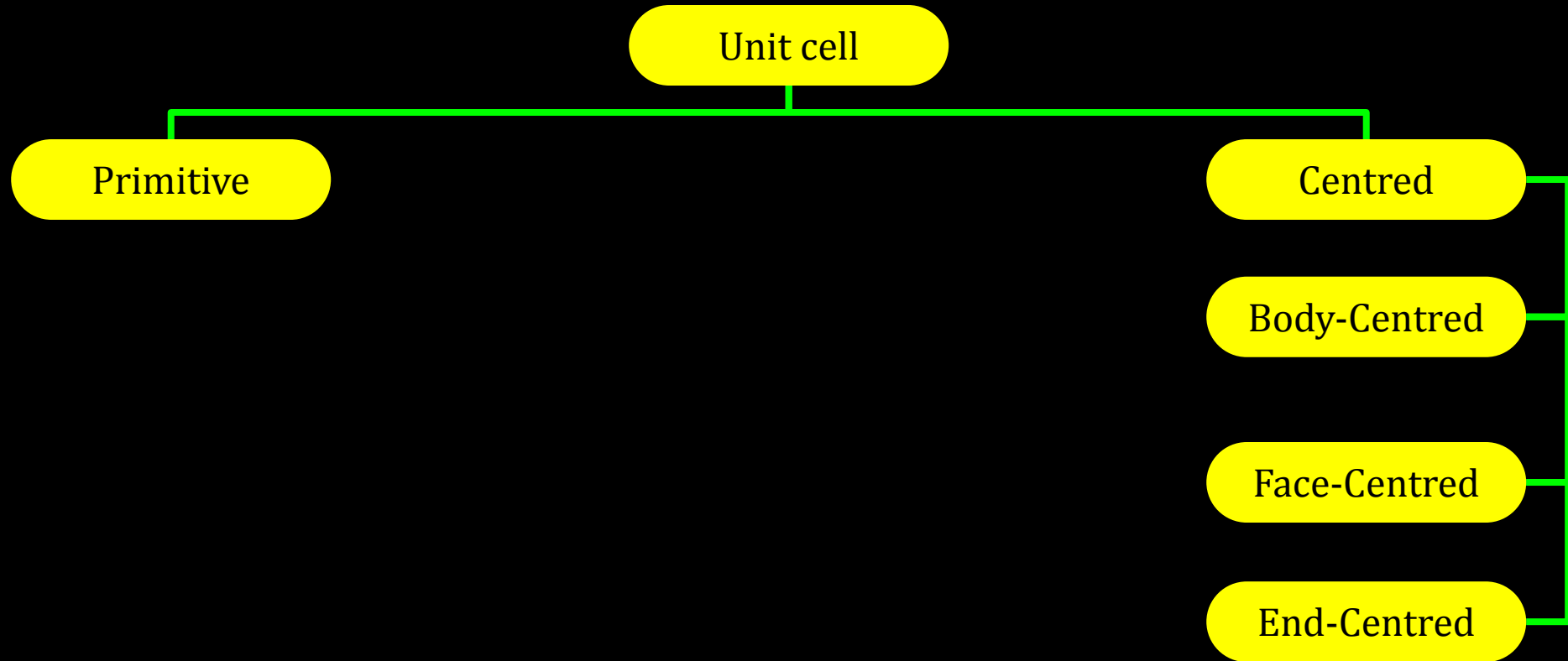
Edge lengths: $a \neq b \neq c$

Axial angles:

$\alpha \neq \beta \neq \gamma \neq 90^\circ$

Examples: $\text{K}_2\text{Cr}_2\text{O}_7$,
 $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, H_3BO_3

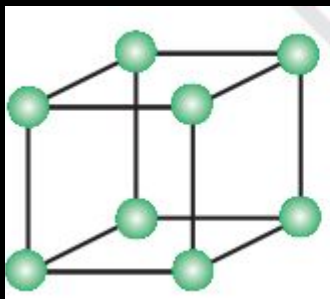




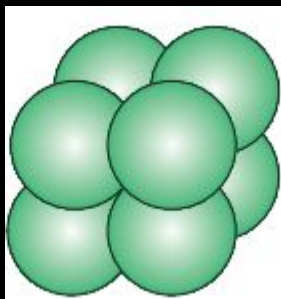
Unit cell

Primitive

Constituent particles are present only on the corner positions of a unit cell.



Open structure



Space-filling



Actual portions of atoms

Centred

Body-Centred

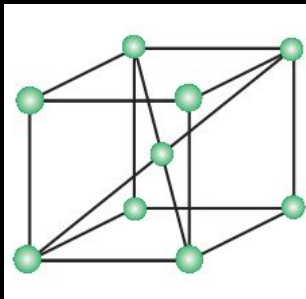
Face-Centred

End-Centred

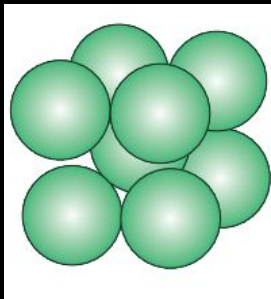
Unit cell

Primitive

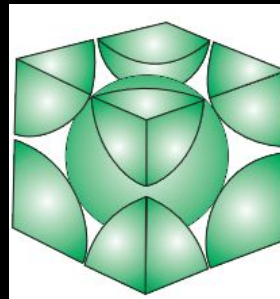
One constituent particle (atom, molecule or ion) at its body-centre besides ones that are at its corners.



Open structure



Space-filling



Actual portions of atoms

Centred

Body-Centred

Face-Centred

End-Centred

Unit cell

Primitive

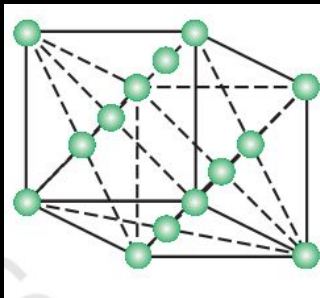
Centred

Contains atoms at all the corners and at the centre of all the faces of the cube

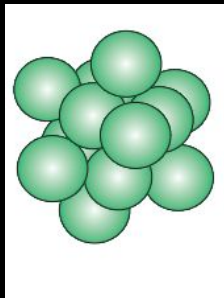
Body-Centred

Face-Centred

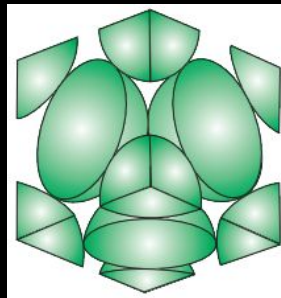
End-Centred



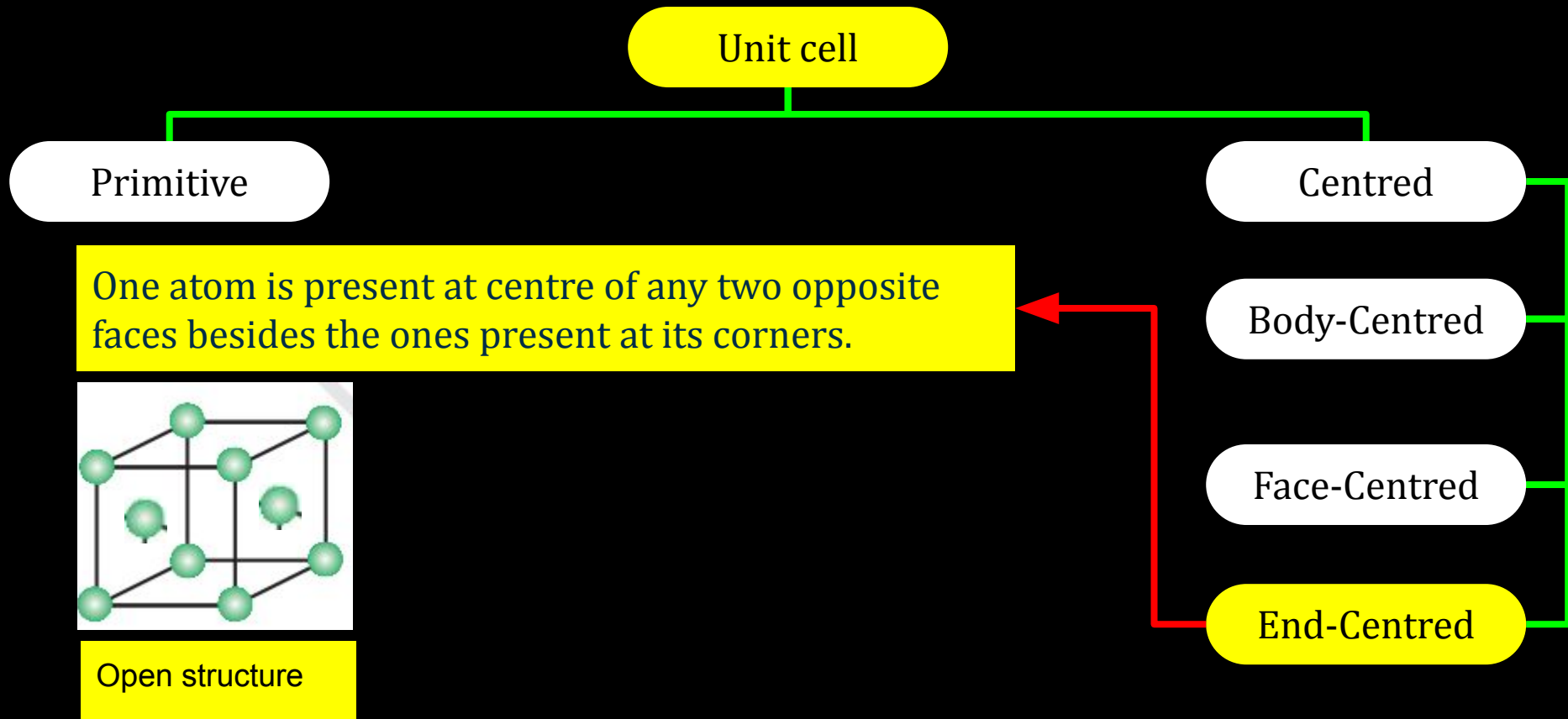
Open structure



Space-filling

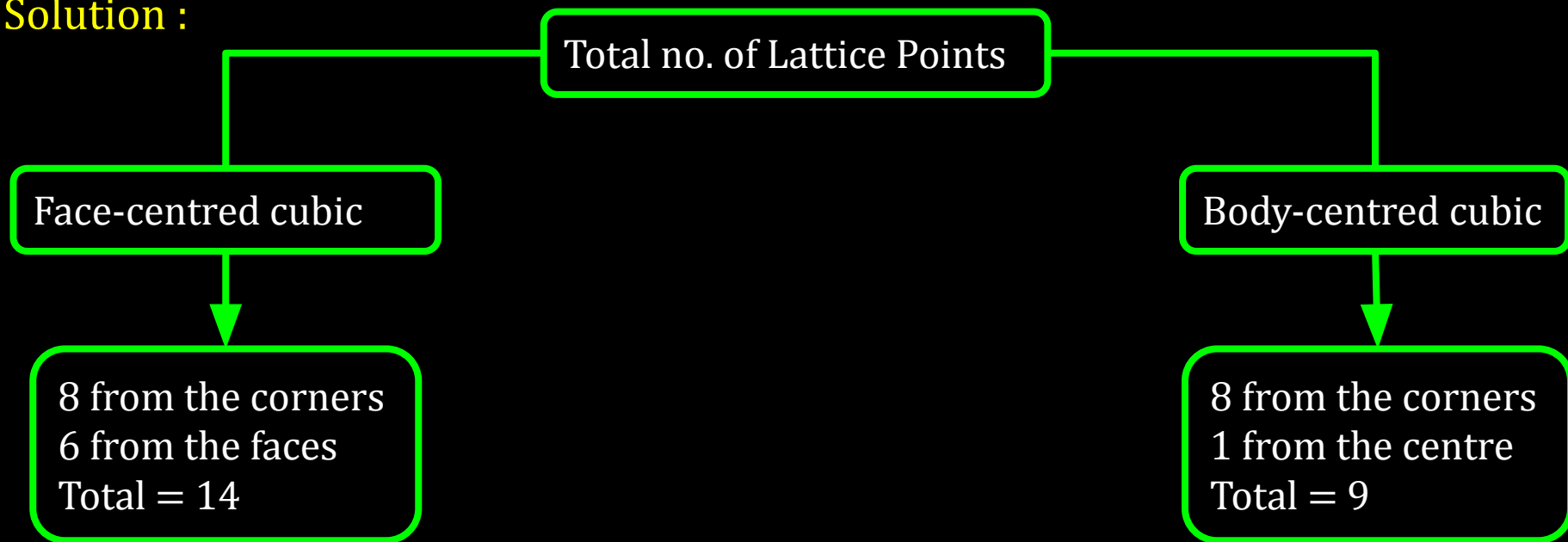


Actual portions of atoms

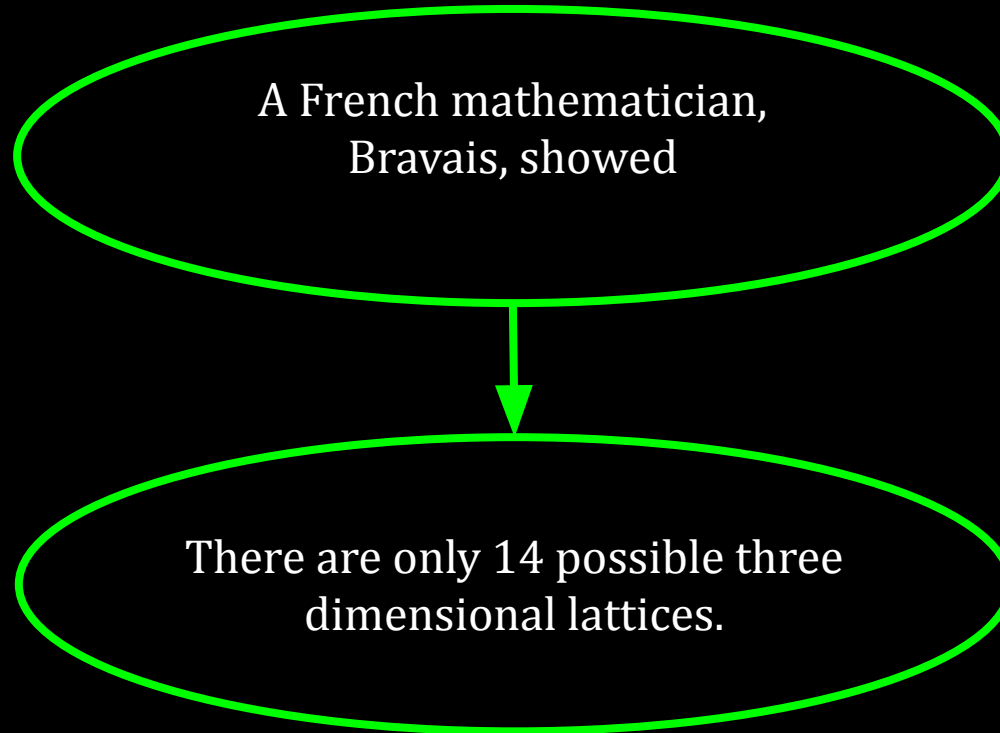


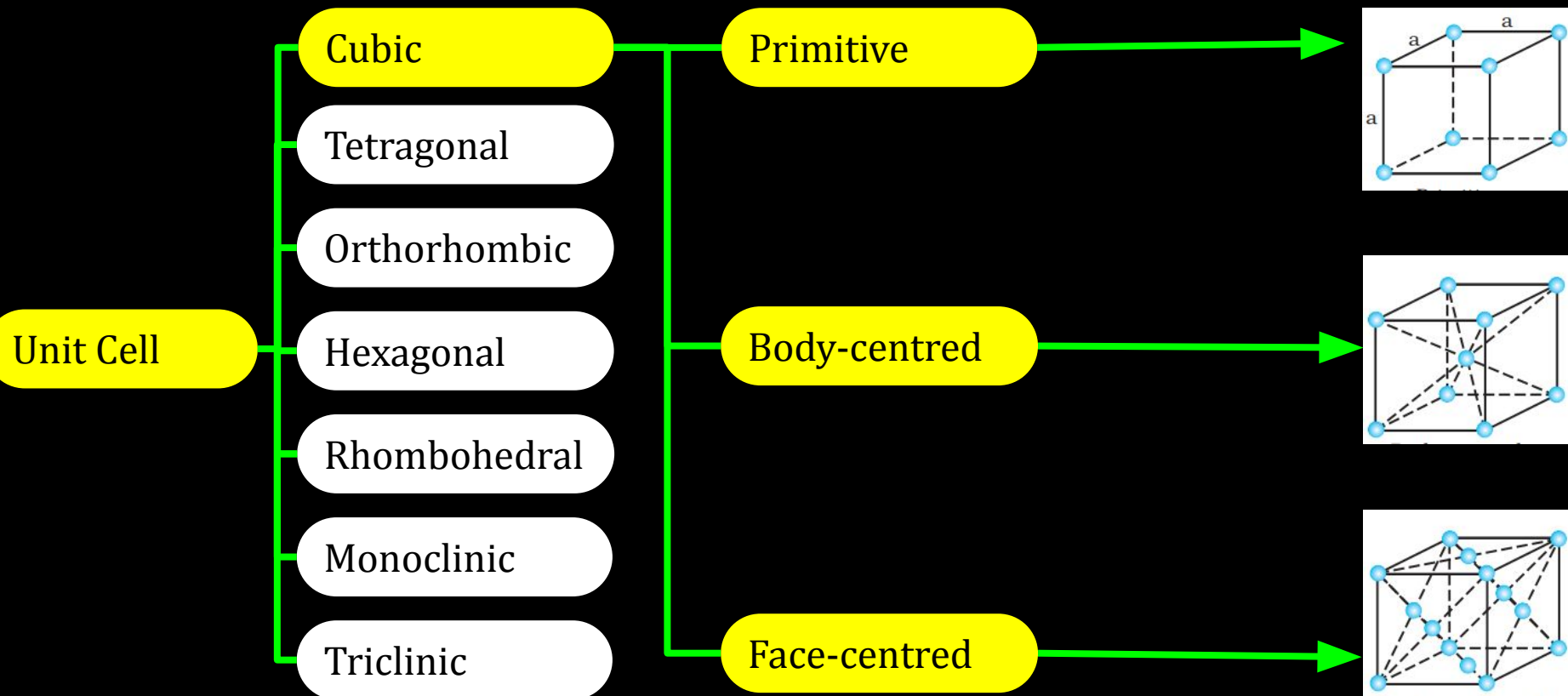
Question : How many lattice points are there in a unit cell of each of the given lattice?
(i) Face-centred cubic (ii) Body-centred cubic

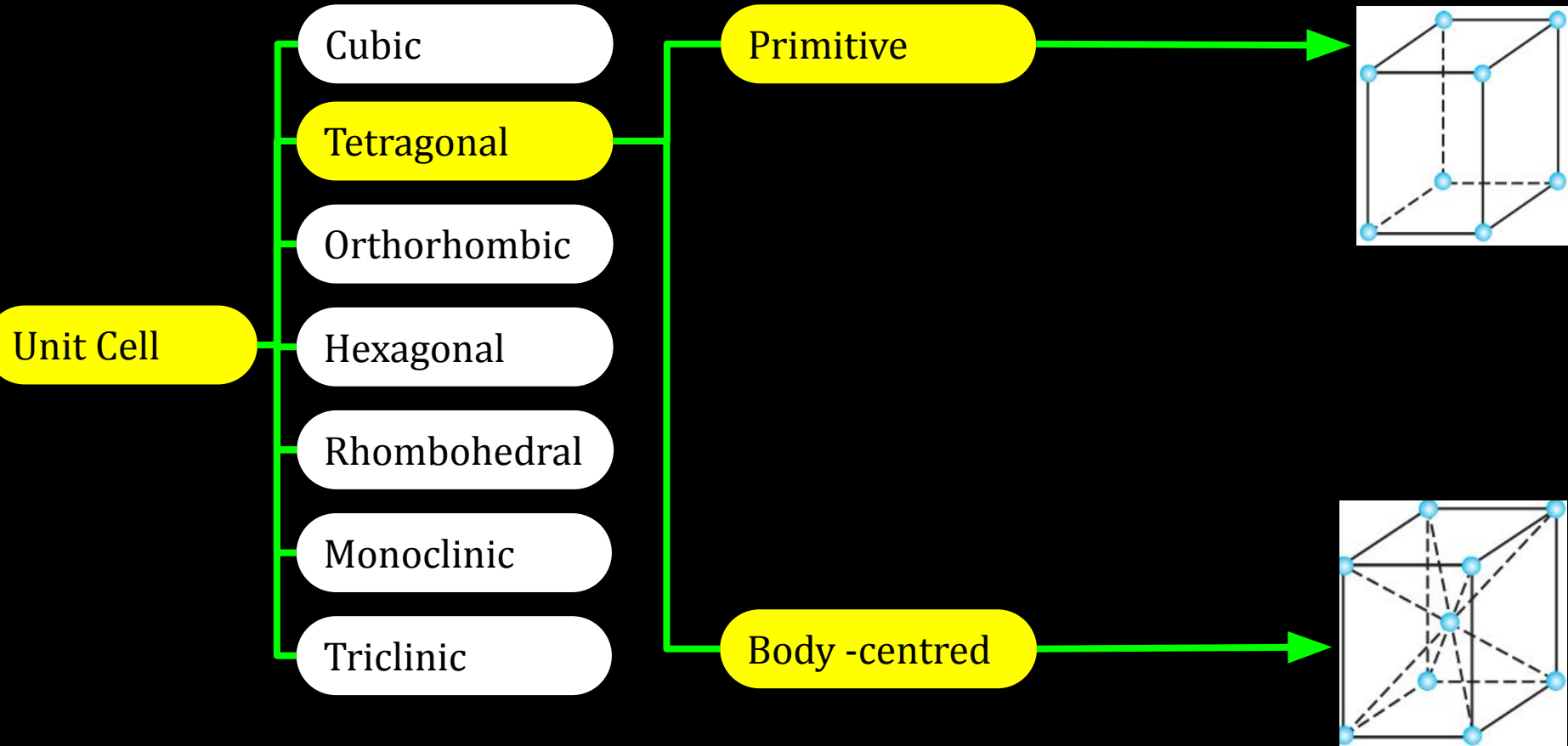
Solution :

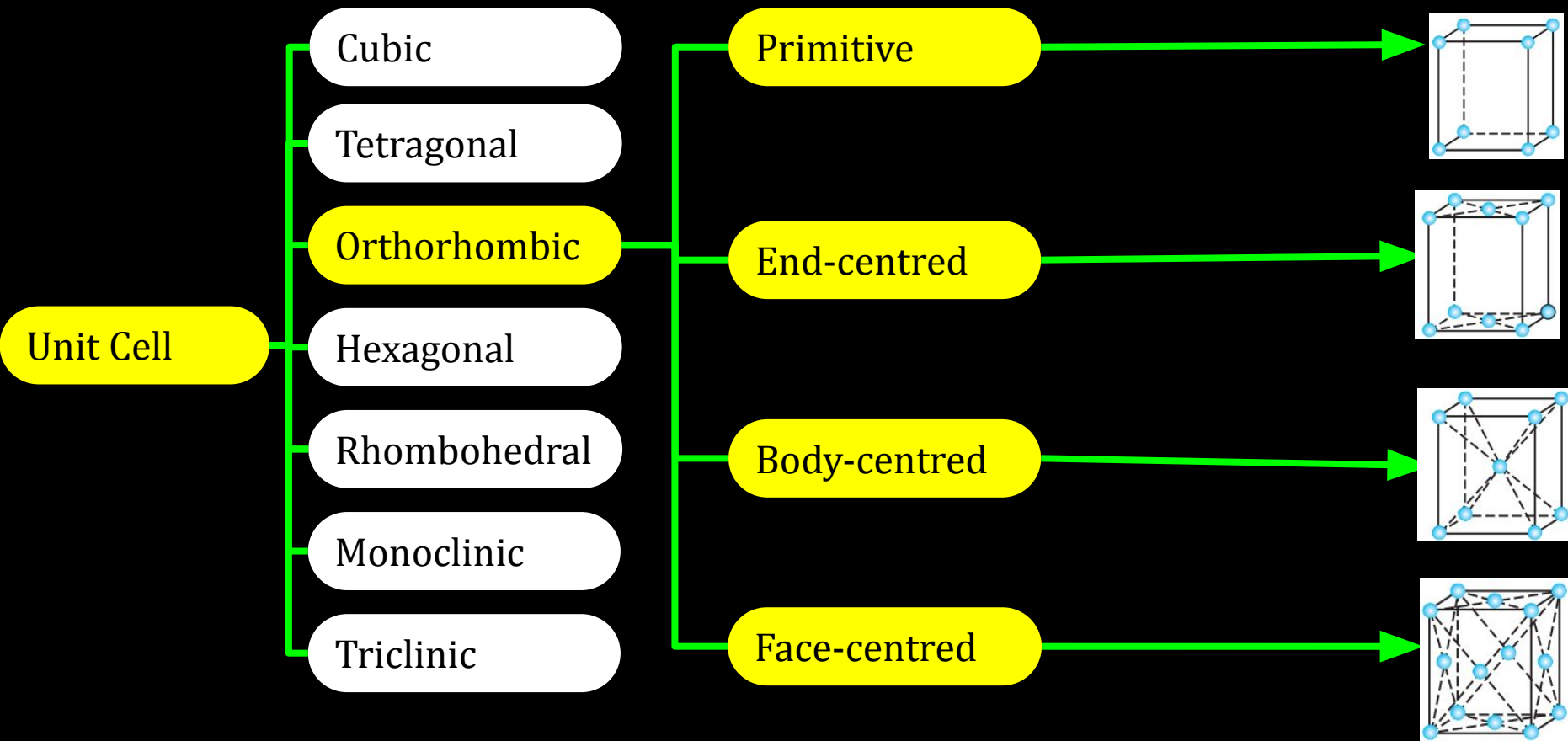


Bravais Lattices:









Unit Cell

Cubic

Tetragonal

Orthorhombic

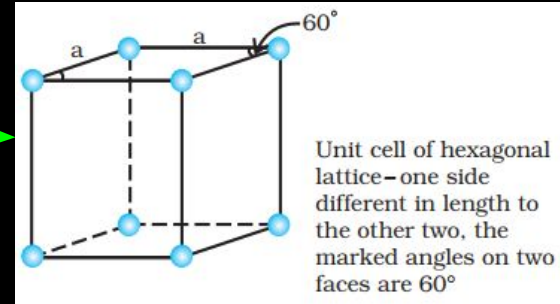
Hexagonal

Rhombohedral

Monoclinic

Triclinic

Primitive



Unit Cell

Cubic

Tetragonal

Orthorhombic

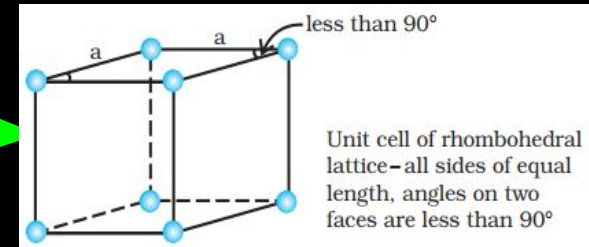
Hexagonal

Rhombohedral

Monoclinic

Triclinic

Primitive



Unit Cell

Cubic

Tetragonal

Orthorhombic

Hexagonal

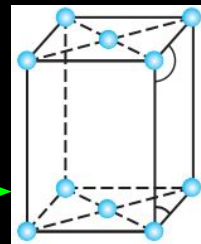
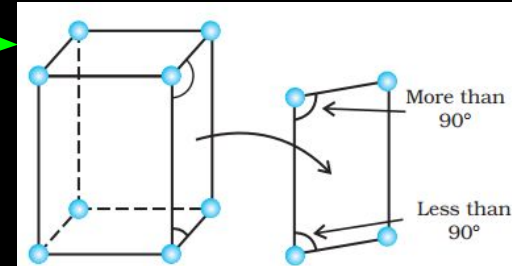
Rhombohedral

Monoclinic

Triclinic

Primitive

End-centred



Unit Cell

Cubic

Tetragonal

Orthorhombic

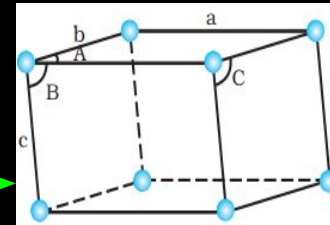
Hexagonal

Rhombohedral

Monoclinic

Triclinic

Primitive



Unit cell of triclinic lattice—
unequal sides a , b , c ,
 A , B , C are unequal
angles with none equal to 90°

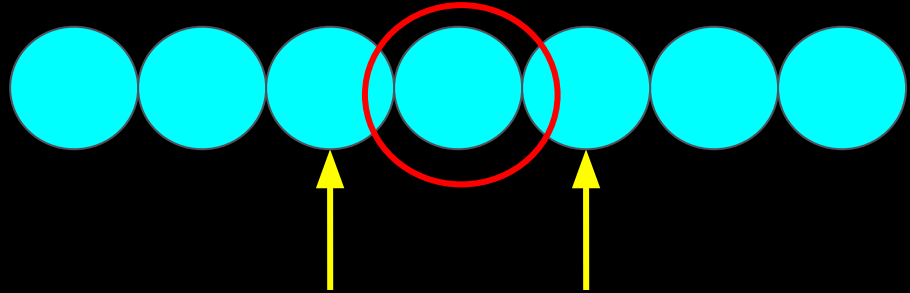
Close Packed Structures

- **Close Packing in One Dimension**
- **Close Packing in Two Dimensions**
- **Close Packing in Three Dimensions**
- **Formula of a Compound and Number of Voids filled**
- **Locating Tetrahedral and Octahedral Voids**

Close Packing in 1D

Contact with 2 neighbours

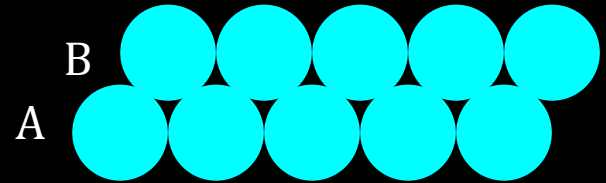
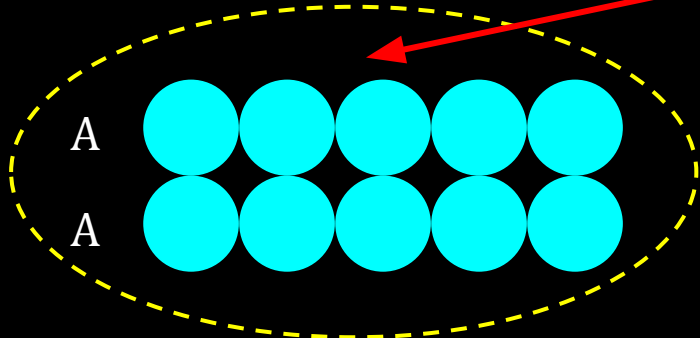
Coordination no. = 2



Close Packing in 2D

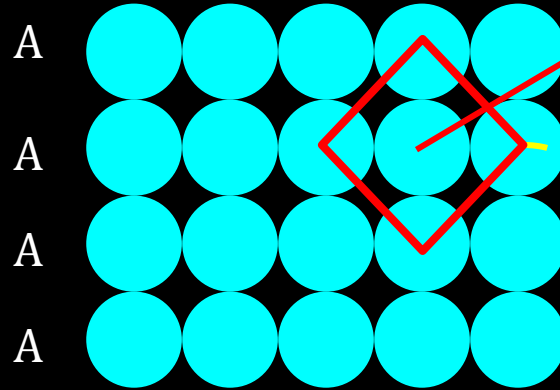
It can be generated by placing the rows of close packed spheres

2 different ways



Spheres are aligned
horizontally & vertically

Contact with 4 neighbours.
coordination no. is = 4

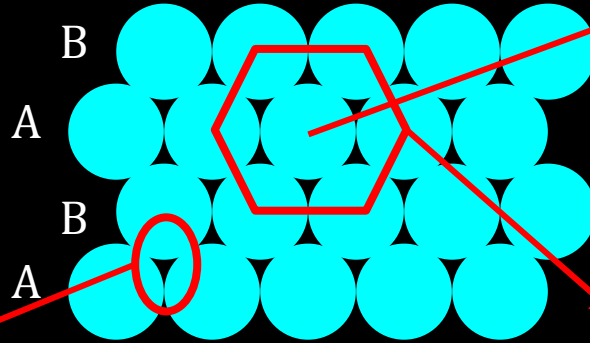


AAA type of arrangement

Square close
packing in 2D

Less free space.
More efficient Packing

Contact with 6 neighbours.
coordination no. = 6



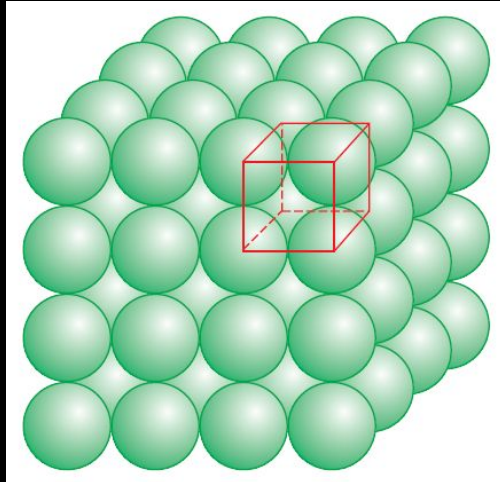
Triangular voids

2-D hexagonal close packing

ABAB type of arrangement

3D close packing from 2D S.C.P. layers

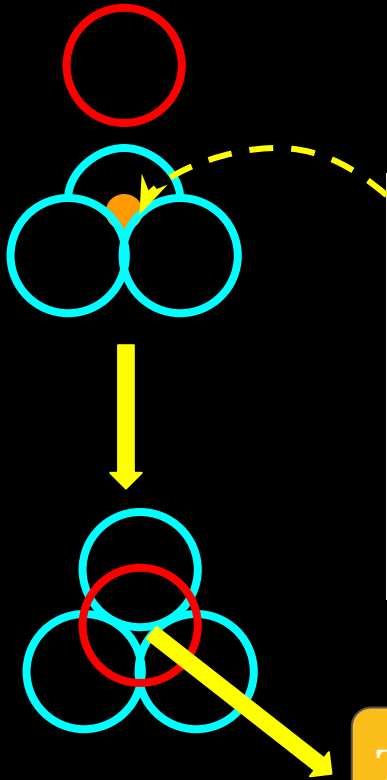
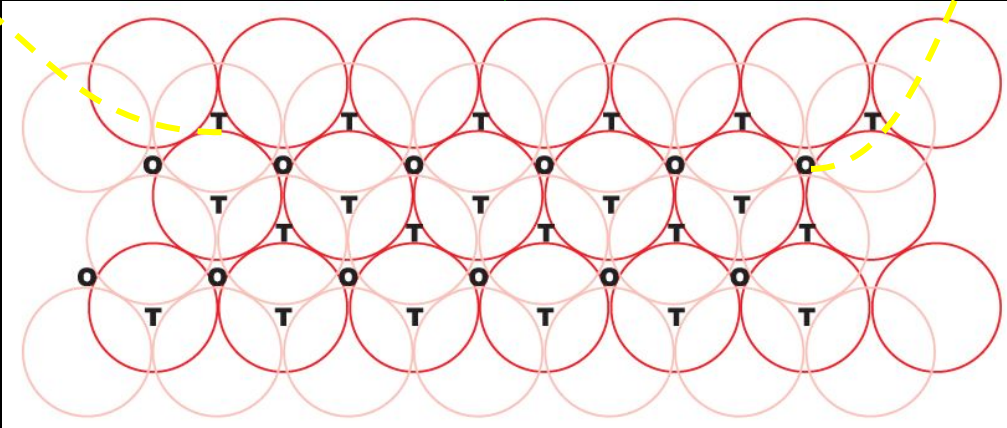
Perfectly aligned Spheres



AAA.. type pattern

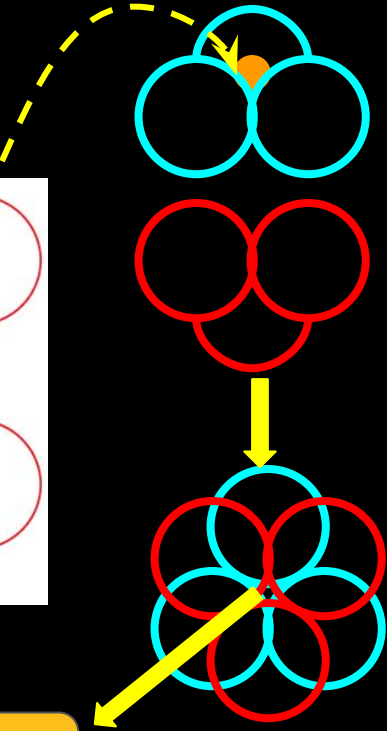
Simple cubic lattice with
primitive cubic unit cell

Tetrahedral and Octahedral voids



Tetrahedral void

Octahedral void



Number of voids

```
graph TD; A[Number of voids] --> B[If the number of close packed spheres be N]; B --> C[Number of octahedral voids = N]; B --> D[Number of tetrahedral voids = 2N];
```

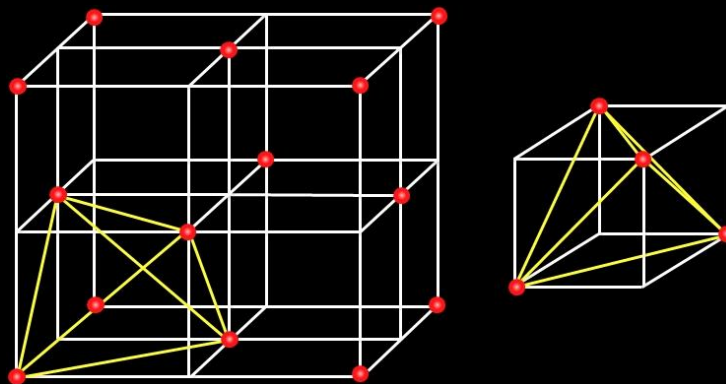
If the number of close packed spheres be N

Number of octahedral voids = N

Number of tetrahedral voids = $2N$

Fcc unit cell $\longrightarrow$ 8 small cubes $\longrightarrow$ Each small cube has 4 atoms

Tetrahedral void in
each small cube = 1
Tetrahedral
voids in unit cell = 8



Regular tetrahedron

No. of atoms = 4
No. of Tetrahedral
voids = 8

ccp or fcc unit cell

Body centre, C is surrounded by 6 atoms on face centres

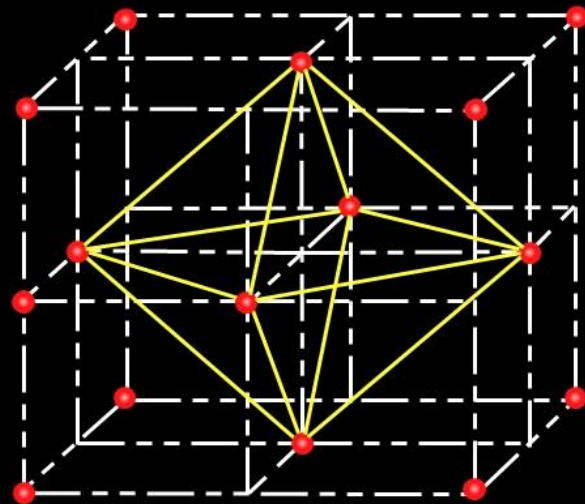
Octahedron

Octahedral void at body centre of cube = 1

Octahedral voids at centre of all the 12 edges = 12

Total Voids = $12 \times \frac{1}{4} + 1 = 4$

Total atoms in ccp unit cell = 4 = Total octahedral voids



ccp or fcc unit cell

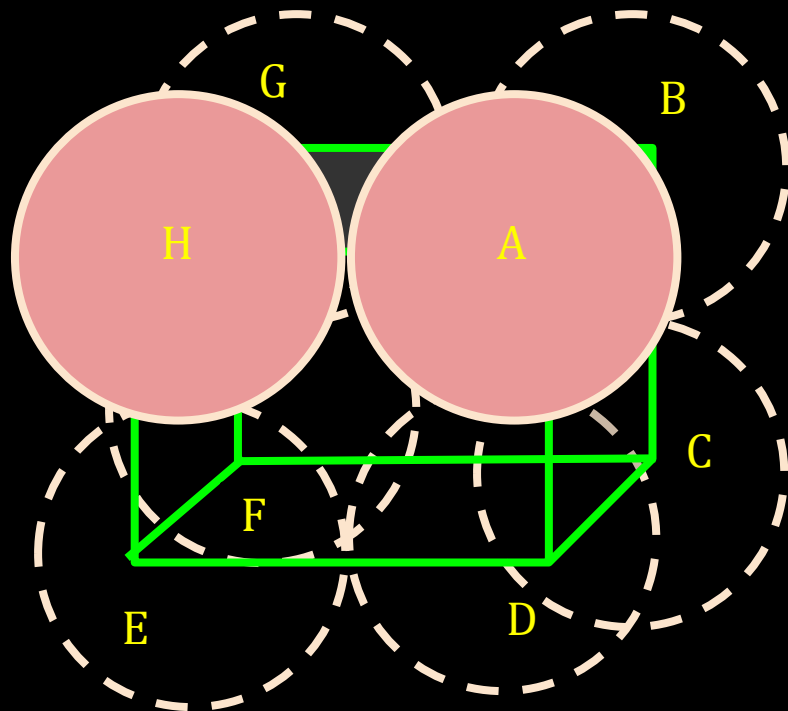
$$\text{Packing Efficiency} = \frac{\text{Volume occupied by atoms in unit cell} \times 100}{\text{Total volume of the unit cell}}$$

P.E. in
hcp and ccp Structures

P.E. in bcc Structures

P.E. in Simple Cubic
Lattice

Packing Efficiency in Simple Cubic Lattice



Coordination Number = 6

Relation b/w edge length (a) and radius(r)
 $a = 2r$

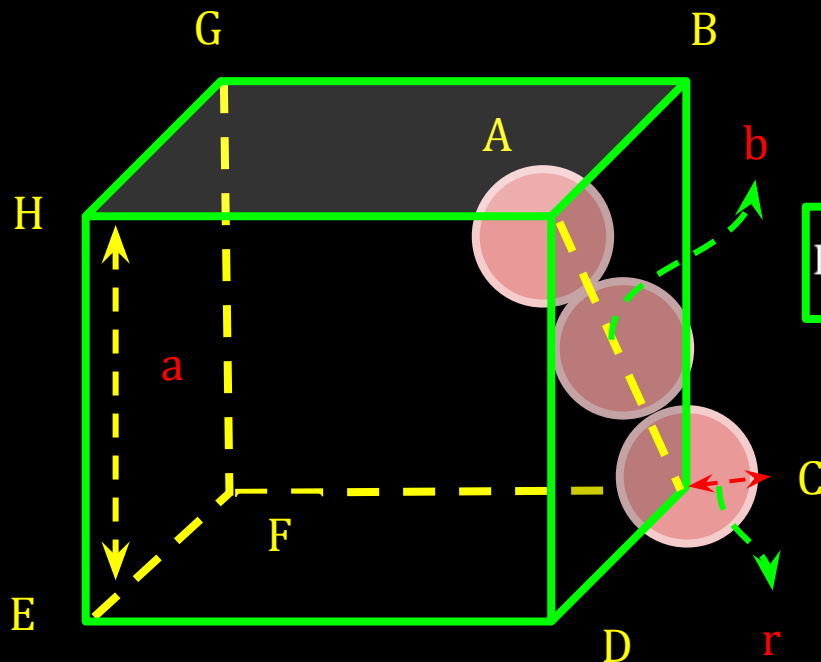
$$\text{P.E} = \frac{\text{Volume occupied by 1 atom} \times 100}{\text{Total volume of the unit cell}}$$

$$\text{P.E} = \frac{(4/3) \pi r^3 \times 100}{8r^3}$$

$$\text{P.E} = 52.4\%$$

Packing Efficiency in hcp and ccp Structures

Coordination Number = 12



Relation b/w edge length (a) and radius(r)
 $a = 2\sqrt{2} r$

$$\text{P.E} = \frac{\text{Volume occupied by 4 spheres in unit cell} \times 100}{\text{Total volume of the unit cell}}$$

$$\text{P.E} = \frac{4 \times (4/3) \pi r^3 \times 100}{(2\sqrt{2} r)^3}$$

$$\text{P.E} = 74\%$$

12C01.4 Packing Efficiency and Crystal Density

Packing Efficiency in bcc Structures

Coordination Number = 8

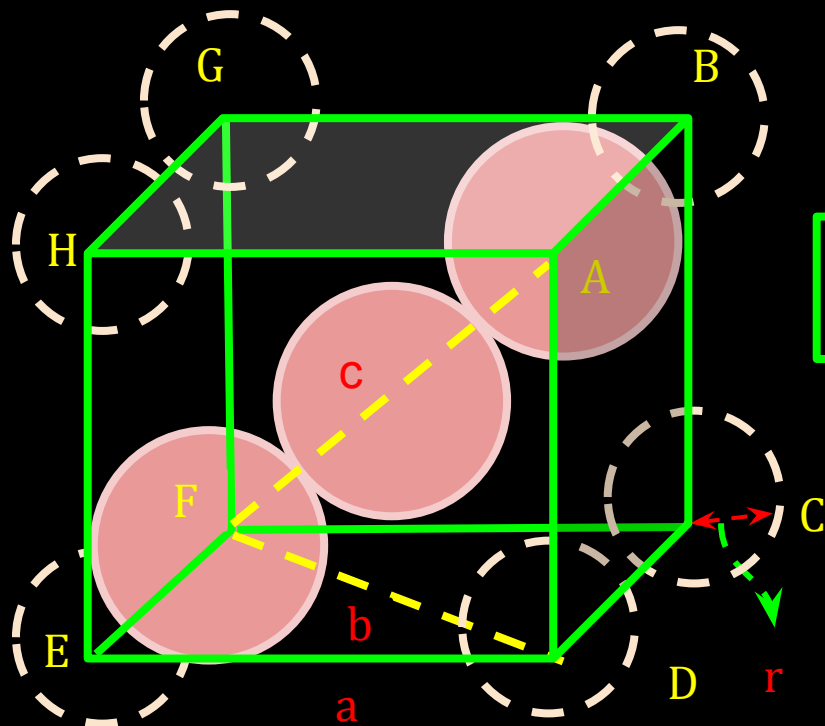
Relation b/w edge length (a) and radius(r)

$$a = 4r/\sqrt{3}$$

$$\text{P.E} = \frac{\text{Volume occupied by 2 spheres in unit cell} \times 100}{\text{Total volume of the unit cell}}$$

$$\text{P.E} = \frac{2 \times (4/3) \pi r^3 \times 100}{[(4/\sqrt{3})r]^3}$$

$$\text{P.E} = 68\%$$

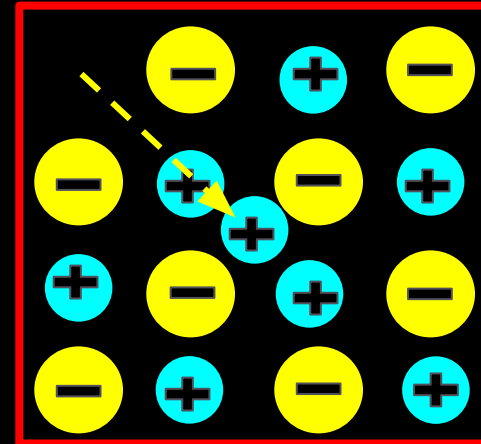


Revision Table

Characteristics	fcc	bcc	Simple Cubic
No. of atoms (Z)	4	2	1
Coordination no.	12	8	6
Relation b/w a and r	$a = 2\sqrt{2} r$	$a = 4r/\sqrt{3}$	$a = 2 r$
P.E	74	68	52.4

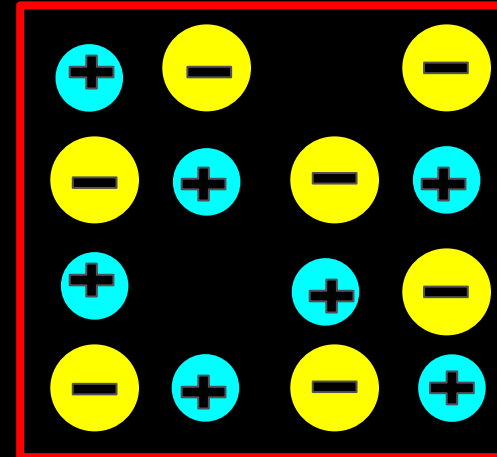
Frenkel Defect

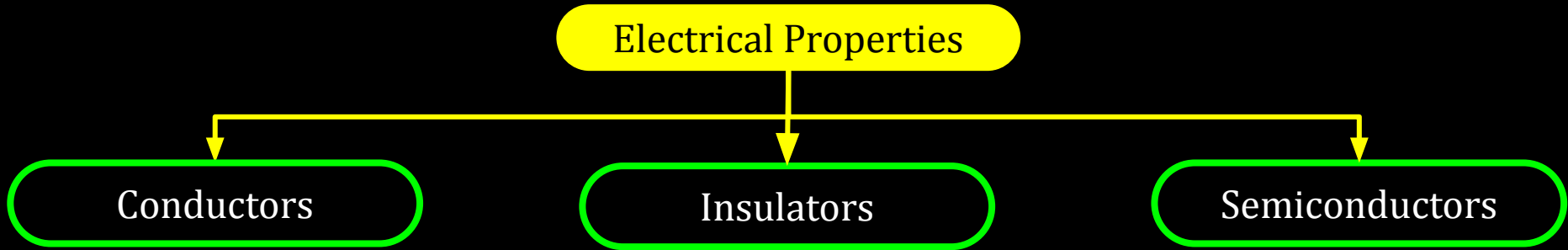
- Smaller ion (cation) moved from normal to interstitial.
- Also called dislocation defect
- Density does not change
- Ex - ZnS, AgCl, AgBr, AgI

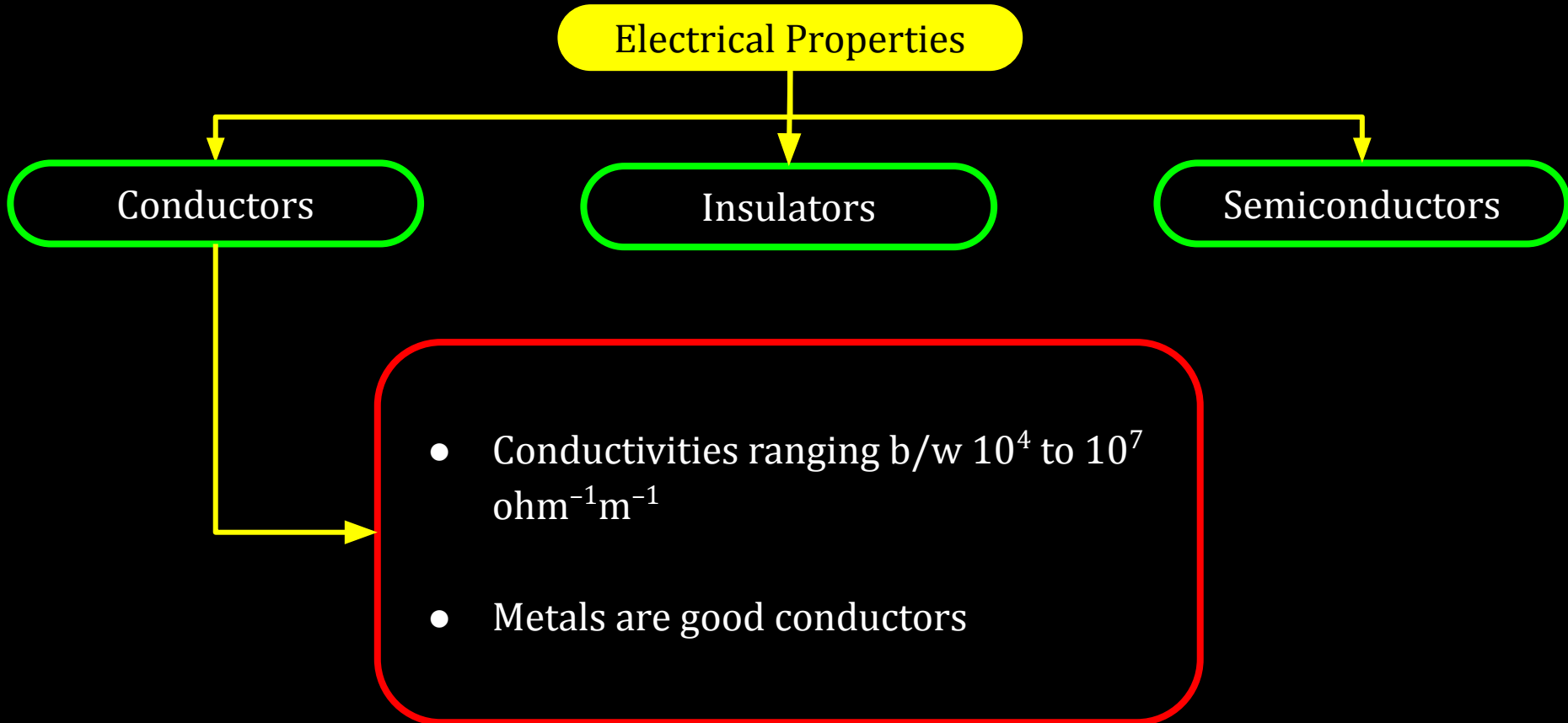


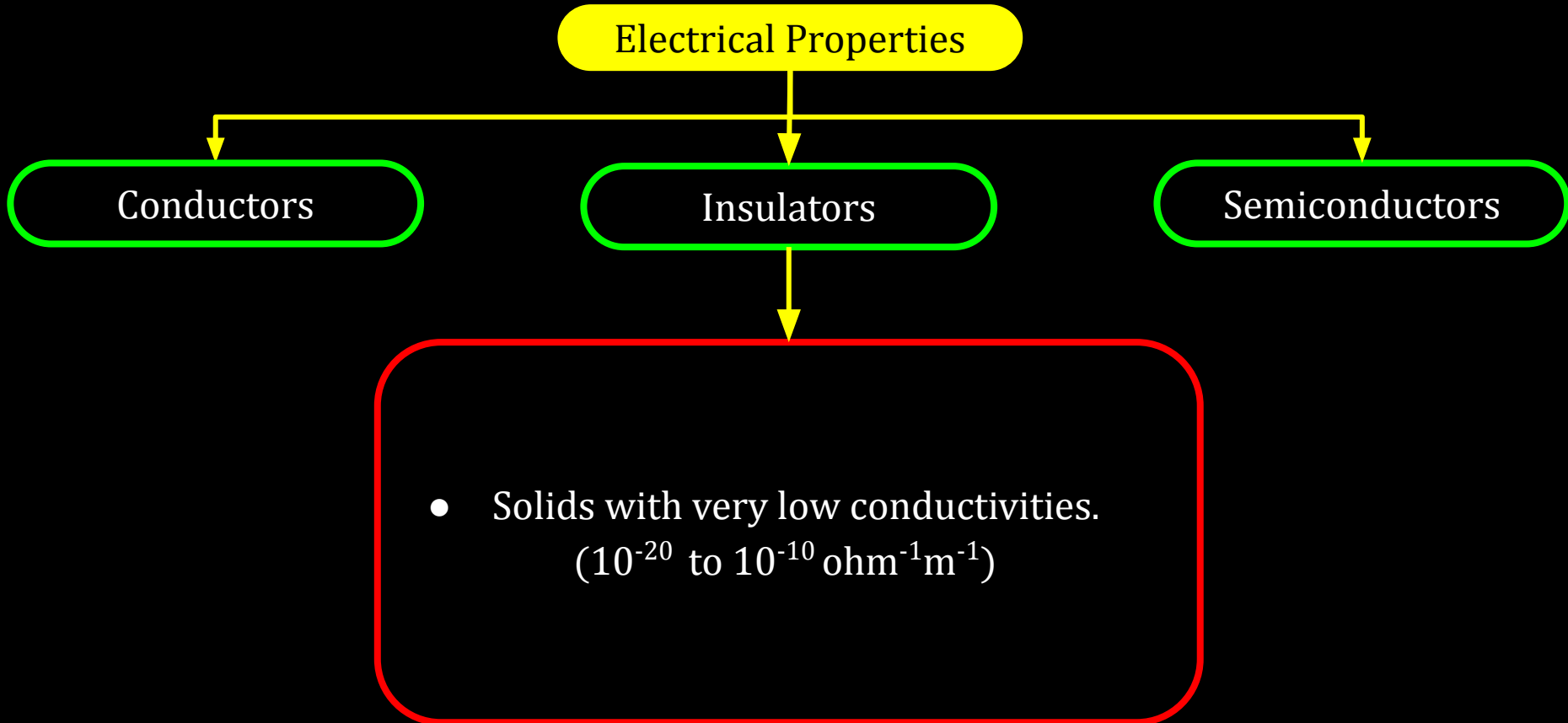
Schottky Defect

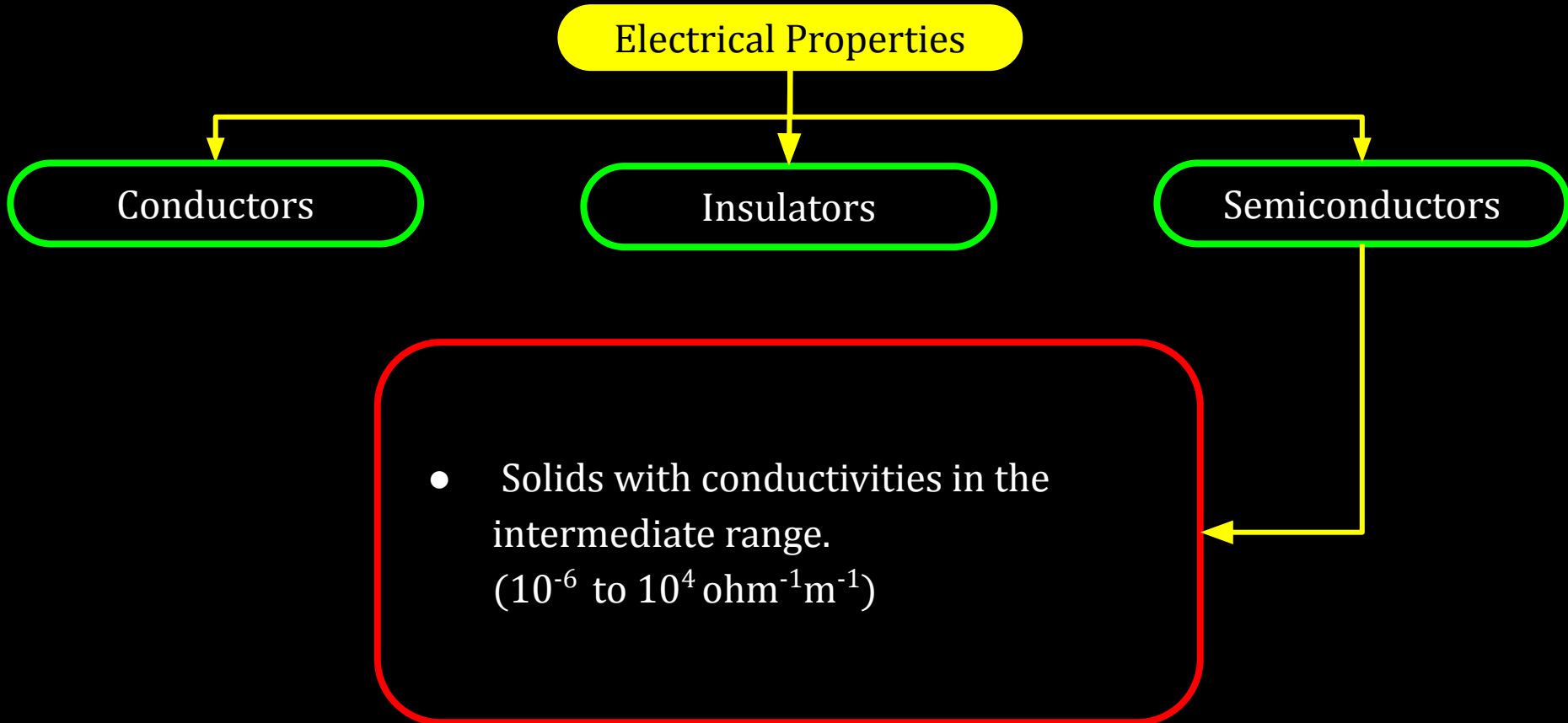
- Number of missing cations and anions are equal.
- Electrical neutral.
- Density of crystal decrease
- Ex. - NaCl, KCl, CsCl, AgBr





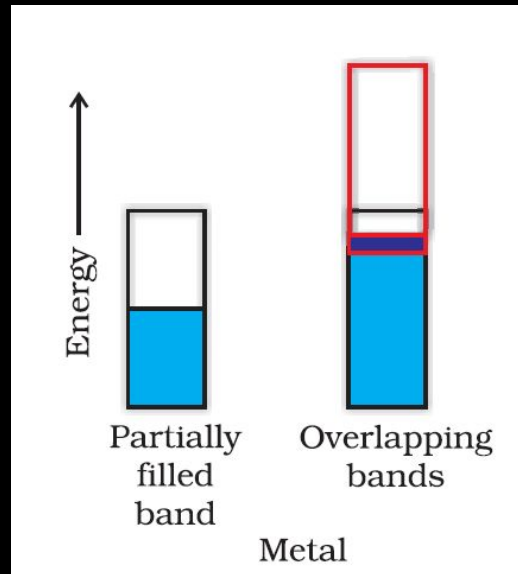






Metals

Conduction Of Electricity in Conductors



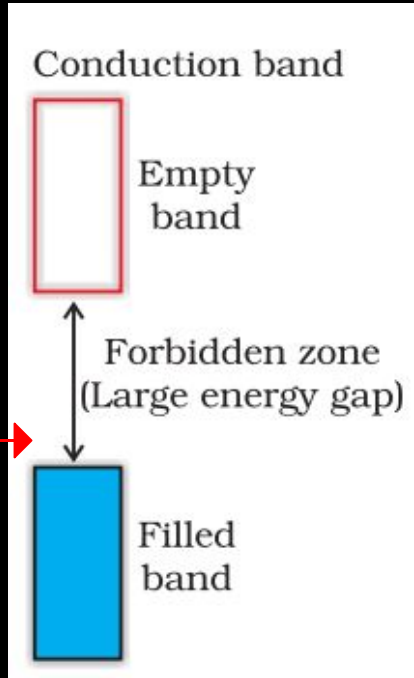
e^- can flow easily under an applied electric field.

Conduction of Electricity in Insulators

Gap between filled valence band & conduction band is large

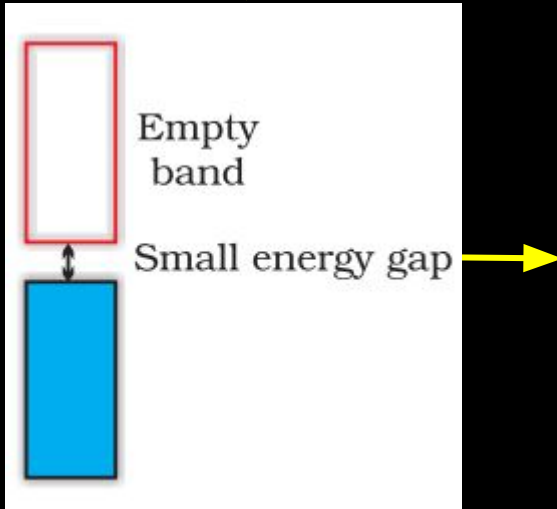
e^- cannot jump.

Very small conductivity.



Conduction of Electricity in Semiconductors

Gap b/w the bands is small



- Conductivity increases with rise in T .
- More electrons can jump.
- These are called intrinsic semiconductors
- Examples are Si and Ge
- Conductivity of these is too low for practical use, and this is increased by doping.

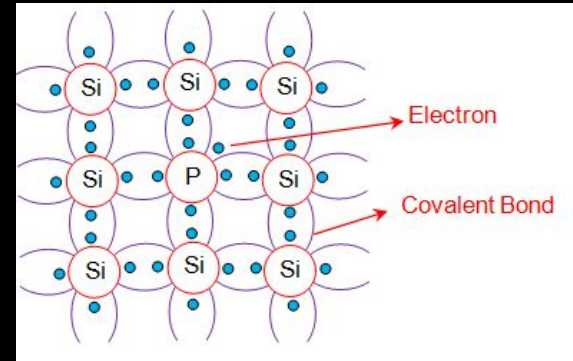
Doping

Process of adding impurity to intrinsic semiconductors to increase their conductivity

Electron – rich impurities

- Group 14 elements like Si and Ge doped with group 15 elements like P
- Delocalised electrons increase the conductivity.

Electron deficient impurities

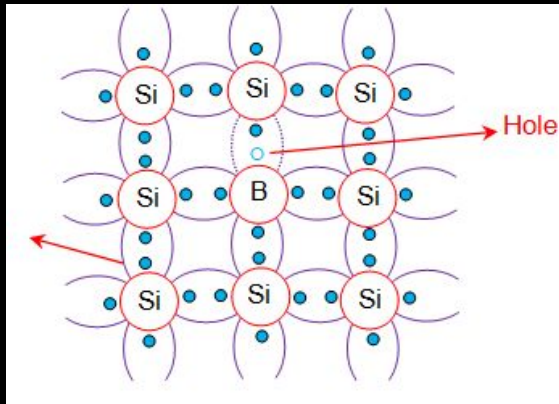


n-type semiconductor

Doping

Process of adding impurity to intrinsic semiconductors to increase their conductivity

Electron – rich impurities



p-type semiconductors

Electron deficient impurities

- Group 14 elements (Si, Ge) doped with group 13 elements like B
- During the movement of e^- from one place to other, electron hole is created at original position